DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(JUNE 2019-NOV 2019)

COMPUTER NETWORKS

V SEMESTER

SYLLABUS

Course Objectives:

1. Given an environment, after analyzing the channel characteristics, appropriate channel access mechanism and data link protocols are chosen to design a network.
2. Given an environment, analyzing the network structure and limitations, appropriate routing protocol is chosen to obtain better throughput.
3. Given various load characteristics and network traffic conditions, decide the transport protocols and timers to be used.
4. Given the requirements of the user, an appropriate Internet protocol and proper security options are chosen.

Course Outcomes:

On successful completion of the module:

1. A student should able to analyze the requirement of various hardware components and software to be developed to establish a network.
2. A student should able to analyze the working conditions of a network and able to provide the solutions to improve the performance of the network.

UNIT - I

Introduction – Uses – Network hardware – software – reference models – example networks – Theoretical basis for communication – transmission media – wireless transmission – Communication satellites.

UNIT - II

Data link layer – design issues – Services - Framing - Error Control - Flow Control - Error detection and correction codes - data link layer protocols -Simplex Protocol – Sliding window Protocols - Medium Access control sublayer – Channel allocation problem – Multiple Access protocols – ALOHA – CSMA Protocols - Collision-Free Protocols - Limited-Contention Protocols - Wireless LANs - 802.11 Architecture - 802.16 Architecture – Data link layer Switching - Uses of Bridges - Learning Bridges - Spanning Tree Bridges - Repeaters, Hubs, Bridges, Switches, Routers, and Gateways - Virtual LANs.

UNIT - III

Network layer – design issues – Routing algorithms - The Optimality Principle - Shortest Path Algorithm – Flooding - Distance Vector Routing - Link State Routing - Hierarchical Routing - Broadcast Routing - Multicast Routing Congestion Control – Approaches - Traffic-Aware Routing - Admission Control - Traffic Throttling - Load Shedding – Internetworking - Tunneling - Internetwork Routing - Packet Fragmentation - IP v4 - IP Addresses – IPv6 - Internet Control Protocols – OSPF – BGP.

UNIT - IV

Transport layer - Services - Berkeley Sockets -Example – Elements of Transport protocols – Addressing - Connection Establishment - Connection Release - Flow Control and Buffering – Multiplexing – Congestion Control - Bandwidth Allocation – Regulating the Sending Rate –UDP-RPC – TCP - TCP Segment Header - Connection Establishment - Connection Release - Transmission Policy - TCP Timer Management - TCP Congestion Control.

UNIT - V

Application Layer – DNS – Name space – Resource records – name servers – e-mail - Architecture and Services - The User Agent - Message Formats - Message Transfer - Final Delivery – WWW – Architecture - Static Web Pages - Dynamic Web Pages and Web Applications - HTTP – Network Security - Introduction to Cryptography - Substitution Ciphers - Transposition Ciphers – Public key algorithms – RSA – Authentication Protocols - Authentication Using Kerberos.

TEXT BOOKS

1. Tanenbaum,A.S. and David J. Wetherall "Computer Networks", 5th ed., Prentice Hall, 2011.

REFERENCE BOOKS

1. Larry L. Peterson and Bruce S. Davie, "Computer Networks- A system approach", 5th edition, ELSEVIER, 2012
2. Stallings, W., 'Data and Computer Communications', 10th Ed., Prentice Hall Int. Ed., 2013
3. James F. Kurose and Keith W. Ross, "Computer Networking: A Top-Down Approach Featuring the Internet", Pearson Education, Third edition, 2006.

WEBSITE

1. http://depa.usst.edu.cn/chenjq/www2/wl/ComputerNetworksTanenbaum.htm
2. http://booksite.mkp.com/9780123850591/lec.php
3. http://williamstallings.com/DataComm/DCC10e-Student/

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SUBJECT NAME: COMPUTER NETWORKS **SUBJECT CODE: CST52**

UNIT - I

Introduction – Uses – Network hardware – software – reference models – example networks – Theoretical basis for communication – transmission media – wireless transmission – Communication satellites.

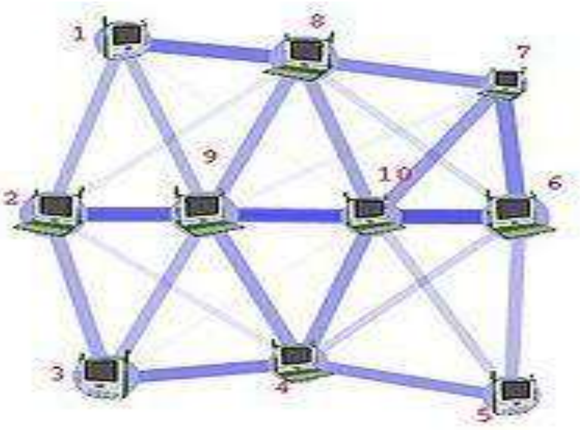
2 MARKS

1. What is Network?

- A network is a group of two or more computer systems linked together.
- Network is an interconnection of systems to share data and information.
- A network is a set of devices (often referred to as **nodes**) connected by communication links. A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

2. Define Computer Network?

A **Computer Network** or **data network** is a telecommunications network which allows computers to exchange data. In computer networks, networked computing devices pass data to each other along network links (data connections). Data is transferred in the form of packets.



3. Define Internet or internetwork?

- A collection of interconnected networks is called an internetwork or internet.
- Internetworking is connecting a computer network with other networks through the use of gateways that provide a common method of routing information packets between the networks. The resulting system of interconnected networks is called an internetwork, or simply an internet.
- Network is an interconnection of systems to share data and information. Internet is network of network or collection of heterogeneous networks.

4. What is distributed system?

A distributed system is a collection of independent computers that appear to the users of the system as a single coherent system.

5. What is meant by Middleware?

- Middleware is computer software that provides services to software applications beyond those available from the operating system.

- Software that acts as a bridge between an operating system or database and applications, especially on a network.

6. What is meant by World Wide Web (WWW)?

World Wide Web (WWW) is a distributed system of inter linked pages that include text, pictures, sound and other information. It enables easy access to the information available on the internet.

WWW is a repository of information (called resource) distributed all over the world, is defined as a world-wide networks of networks.

7. Mention the uses of Computer Networks?

The various uses of Computer Networks are

1. Business Applications
2. Home Applications
3. Mobile Users
4. Social Issues

8. What is meant by Virtual Private Networks (VPN)?

Virtual Private Networks (VPN) is a network that is constructed by using public wires (usually the Internet) to connect to a private network, such as a company's internal network. There are a number of systems that enable you to create networks using the Internet as the medium for transporting data.

9. What is meant by Client?

The client part of a Client-Server architecture. Typically, a client is an application that runs on a personal computer or workstation and relies on a server to perform some operations. For example, an e-mail client is an application that enables you to send and receive e-mail.

10. What is meant by Server? List its types?

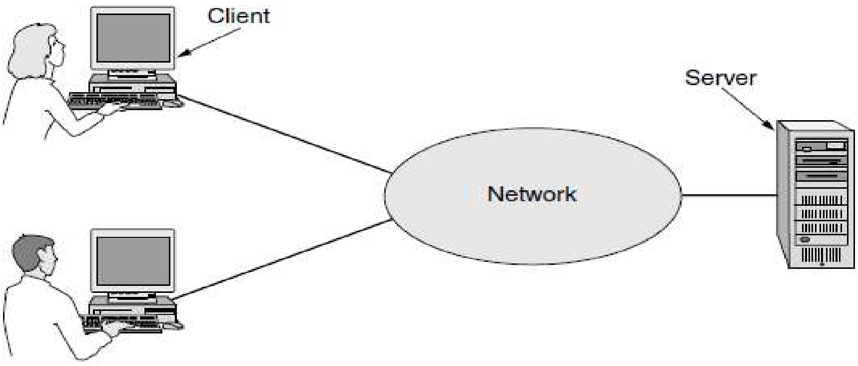
The data are stored on powerful computers called servers. A computer or device on a network that manages network resources.

There are many different types of servers.

- **File server**: a computer and storage device dedicated to storing files. Any user on the network can store files on the server.
- **Print server:** a computer that manages one or more printers, and a network server is a computer that manages network traffic.
- **Database server:** a computer system that processes database queries.

11. What is Client-Server model?

The client-server model is widely used and forms the basis of much network usage. The most popular realization is that of a Web application, in which the server generates Web pages based on its database in response to client requests that may update the database. The client-server model is applicable when the client and server are both in the same building.



12. What is meant by IP telephony?

- IP telephony (Internet Protocol telephony) is a general term for the technologies that use the Internet Protocol's packet-switched connections to exchange voice, fax, and other forms of information that have traditionally been carried over the dedicated circuit-switched connections of the public switched telephone network (PSTN).
- Telephone calls between employees may be carried by the computer network instead of by the phone company. This technology is called **IP telephony** or **Voice over IP (VoIP)** when Internet technology is used.

13. What is meant by Cookies?

A cookie is a mechanism that allows the server to store its own information about a user on the user's own computer. You can view the cookies that have been stored on your hard disk. The location of the cookies depends on the browser.

A small files called **cookies** that Web browsers store on users' computers allow companies to track users' activities in cyberspace and may also allow credit card numbers, social security numbers, and other confidential information to leak all over the Internet.

14. What is meant by Phishing?

- **Phishing** messages masquerade as originating from a trustworthy party, for example, your bank, to try to trick you into revealing sensitive information, for example, credit card numbers. Identity theft is becoming a serious problem as thieves collect enough information
- Phishing is an e-mail fraud method in which the perpetrator sends out legitimate-looking email in an attempt to gather personal and financial information from recipients.

15. Name the two types of transmission technology? **(MAY 2016, 2017)**

There are two types of transmission technology that are in widespread use:

1. broadcast links and
2. point-to-point links

16. What is broadcast links and point-to-point links?

- Point-to-point links connect individual pairs of machines. To go from the source to the destination on a network made up of point-to-point links, short messages, called packets in certain contexts, may have to first visit one or more intermediate machines.
- On a broadcast network, the communication channel is shared by all the machines on the network; packets sent by any machine are received by all the others.

17. What is meant by unicasting, broadcasting and multicasting?

- Point-to-point transmission with exactly one sender and exactly one receiver is sometimes called **unicasting.**
- When a packet with this code is transmitted, it is received and processed by every machine on the network. This mode of operation is called **broadcasting.**
- Broadcast systems also support transmission to a subset (group) of the machines, which known as **multicasting.**

18. What is meant by PAN, LAN, MAN and WAN?

- A Personal Area Networks (PAN) let devices communicate over the range of a person. A common example is a wireless network that connects a computer with its peripherals.
- A Local Area Networks (LAN) is a privately owned network that operates within and nearby a single building like a home, office or factory. LANs are widely used to connect personal computers and consumer electronics to let them share resources (e.g., printers) and exchange information. When LANs are used by companies, they are called enterprise networks.
- A Metropolitan Area Network (MAN) is a network with a size between a LAN and a WAN. It normally covers the area inside a town or a city. It is designed for customers who need a high-speed connectivity, normally to the Internet, and have endpoints spread over a city or part of city.
- A Metropolitan Area Network (MAN) covers a city. The best-known examples of MANs are the cable television networks available in many cities.
- A Wide Area Network (WAN) provides long-distance transmission of data, image, audio, and video information over large geographic areas that may comprise a country, a continent, or even the whole world.

19. What is meant by Access Point (AP)?

- In computer networking, a wireless access point (AP) is a device that allows wireless devices to connect to a wired network using Wi-Fi, or related standards.
- This device, called an AP (Access Point), wireless router, or base station, relays packets between the wireless computers and also between them and the Internet.

20. Define Ethernet?

A system for connecting a number of computer systems to form a local area network, with protocols to control the passing of information and to avoid simultaneous transmission by two or more systems.

21. What is the purpose of network interface card? (NOV 2016)

- Each station on an Ethernet network (such as a PC, workstation, or printer) has its own network interface card (NIC).
- The NIC fits inside the station and provides the station with a 6-byte physical address.
- A computer uses a network interface card (NIC) to become part of a network.
- The NIC contains the electronic circuitry required to communicate using a wired connection (e.g., Ethernet) or a wireless connection (e.g., WiFi).
- A network interface card is also known as a network interface controller, network adapter, or Local Area Network (LAN) adapter.

22. Define Switch and Router?

- Switching elements, or just switches, are specialized computers that connect two or more transmission lines. When data arrive on an incoming line, the switching element must choose an outgoing line on which to forward them.
- A router is a device that forwards data packets along networks. A router is connected to at least two networks, commonly two LANs or WANs or a LAN and its ISP's network. Routers are located at gateways, the places where two or more networks connect.

23. Define Protocol?

A protocol is a set of rules that govern data communications. A protocol defines what is communicated, how it is communicated, and when it is communicated. The key elements of a protocol are syntax, semantics, and timing.

24. What is Network Architecture?

A set of layers and protocols is called a network architecture. The specification of an architecture must contain enough information to allow an implementer to write the program or build the hardware for each layer so that it will correctly obey the appropriate protocol.

25. Define Protocol Stack?

A list of the protocols used by a certain system, one protocol per layer, is called a protocol stack.

26. What is meant by error detection and error correction?

- One mechanism for finding errors in received information uses codes for **error detection**. Information that is incorrectly received can then be retransmitted until it is received correctly.

- More powerful codes allow for **error correction**, where the correct message is recovered from the possibly incorrect bits that were originally received.

27. What is meant by statistical multiplexing?

Many designs share network bandwidth dynamically, according to the short term needs of hosts, rather than by giving each host a fixed fraction of the bandwidth that it may or may not use. This design is called **statistical multiplexing**, meaning sharing based on the statistics of demand.

28. List the design issues to secure networks?

The design issue is to secure the network by defending it against different kinds of threats are

- Confidentiality
- Authentication
- Integrity

29. Define Packets?

A piece of a message transmitted over a packet-switching network. One of the key features of a packet is that it contains the destination address in addition to the data. In IP networks, packets are often called **datagrams.**

30. Difference between Connection-oriented and Connection-less Service? (MAY 2017)

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Movie download
	Unreliable connection	Voice over IP
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Text messaging
	Request-reply	Database query

31. Difference between store-and-forward switching and cut-through switching?

- When the intermediate nodes receive a message in full before sending it on to the next node, this is called **store-and-forward switching.**
- The alternative, in which the onward transmission of a message at a node starts before it is completely received by the node, is called **cut-through switching.**

- Normally, when two messages are sent to the same destination, the first one sent will be the first one to arrive. However, it is possible that the first one sent can be delayed so that the second one arrives first.

32. List the various service primitives?

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
ACCEPT	Accept an incoming connection from a peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

33. Define Internet Protocol (IP)?

The Internet Protocol (IP) is the principal communications protocol in the Internet protocol suite for relaying datagrams across network boundaries. Its routing function enables internetworking, and essentially establishes the Internet.

34. Difference between OSI Model and TCP/IP Model?

OSI Model	TCP/IP Model
OSI stands for Open System Interconnection because it allows any two different systems to communicate regardless of their architecture.	TP/IP stands for Transmission Control Protocol/Internet Protocol. It is named after these protocols, being part of this model.
OSI model has seven layers.	TCP/IP has four layers.
This model provides clear distinction between services, interfaces and protocols	It does not clearly distinguish between services, interfaces & protocols.
In this model, Protocols do not fit well into the model.	TCP and IP protocols fit well in the model.
OSI model supports both connection oriented & connectionless in network layer but connection oriented comm. In transport layer.	TCP/IP supports only connectionless comm. In network layer but supports both in transport layer.

35. What are the functions of the transport layer in the OSI reference model? (NOV 2015)

- Service-point addressing
- Segmentation and reassembly
- Connection control
- Flow control
- Error control

36. What is meant by TCP?

TCP (Transmission Control Protocol), is a reliable connection-oriented protocol that allows a byte stream originating on one machine to be delivered without error on any other machine in the internet. It segments the incoming byte stream into discrete messages and passes each one on to the internet layer. At the destination, the receiving TCP process reassembles the received messages into the output stream

37. What is meant by UDP?

UDP (User Datagram Protocol), is an unreliable, connectionless protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own. It is also widely used for one-shot, client-server-type request-reply queries and applications in which prompt delivery is more important than accurate delivery, such as transmitting speech or video.

38. List the critical issues in OSI Model?

- Bad timing
- Bad technology
- Bad implementations
- Bad politics

39. List the examples of computer networks?

1. The Internet
2. Third-Generation Mobile Phone Networks
3. Wireless LANs: 802.11
4. RFID and Sensor Networks

40. Define Sockets?

A socket is one endpoint of a two-way communication link between two programs running on the network. A socket is bound to a port number so that the TCP layer can identify the application that data is destined to be sent to. An endpoint is a combination of an IP address and a port number**.**

41. Define multipath Fading?

Multipath signals are received in a terrestrial environment, i.e., where different forms of propagation are present and the signals arrive at the receiver from transmitter via a variety of paths. Therefore there would be multipath interference, causing multi- path fading.

42. List the various types of RFID?

- Passive RFID
- Active RFID
- UHF RFID (Ultra-High Frequency RFID)

- HF RFID (High Frequency RFID)
- LF RFID (Low Frequency RFID)

43. List the various applications in sensors?

- monitoring bird habitats,
- volcanic activity,
- business applications including healthcare,
- monitoring equipment for vibration, and
- tracking of frozen,
- refrigerated

44. Define bandwidth?

The bandwidth of a composite signal is the difference between the highest and the lowest frequencies contained in that signal.

45. What is meant by Nyquist Bit Rate?

For a noiseless channel, the Nyquist bit rate formula defines the theoretical maximum bit rate

BitRate = 2 x bandwidth x 10g2 L

In this formula, bandwidth is the bandwidth of the channel, L is the number of signal levels used to represent data, and BitRate is the bit rate in bits per second.

46. What is meant by Shannon capacity?

Claude Shannon introduced a formula, called the Shannon capacity, to determine the theoretical highest data rate for a noisy channel:

Capacity =bandwidth X log2 (1 +SNR)

In this formula, bandwidth is the bandwidth of the channel, SNR is the signal-to-noise ratio, and capacity is the capacity of the channel in bits per second.

47. What are Half and Full duplex mode?

- In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa.
- In full-duplex mode, both stations can transmit and receive simultaneously.

48. List the various types of network topology?

- Mesh
- Star
- Ring
- Bus

49. What are the two categories of physical media?

1. Guided Media
    - Magnetic Media
    - Twisted –Pair cable
        - Shielded Twisted Pair cable
        - Unshielded Twisted Pair cable
    - Coaxial Cable
    - Fiber-optic cable
    - Power lines
2. Unguided Media
    - Terrestrial microwave
    - Satellite Communication

50. Differentiate guided and unguided transmission medium? (NOV 2016)

Guided transmission medium	Unguided transmission medium
Guided media, which are those that provide a conduit from one device to another.	Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication.
A signal traveling along any of these media is directed and contained by the physical limits of the medium.	Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.
Guided media include twisted-pair cable, coaxial cable, and fiber-optic cable.	Unguided media include Radio wave, Micro wave and Infrared.

51. What is meant by twisted pair cable?

- Twisted pair consists of two insulated copper wires, typically about 1 mm thick. A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together.
- One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference.
- The most common application of the twisted pair is the telephone system.

52. What are the two types of Fibers?

- Single mode Fiber
- Multi-mode Fiber

53. Comparison between comparison of semiconductor diodes and LEDs?

Item	LED	Semiconductor laser
Data rate	Low	High
Fiber type	Multi-mode	Multi-mode or single-mode
Distance	Short	Long
Lifetime	Long life	Short life
Temperature sensitivity	Minor	Substantial
Cost	Low cost	Expensive

54. Mention the advantages of Optical fiber?

- Higher bandwidth
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials
- Light weight
- Greater immunity to tapping

55. Mention the disadvantages of Optical fiber?

- Installation and maintenance
- Unidirectional light propagation
- Cost

56. What is unguided media?

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as wireless communication. Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

57. What is the difference between a passive star and an active repeater in a fiber optic network? (MAY 2016)

- A passive star has no electronics. The light from one fiber illuminates a number of others.
- The passive star combines all the incoming signals and transmits the merged result out on all lines.
- An active repeater converts the optical signal to an electrical one for further processing.
- The active repeater is an incoming light is converted to an electrical signal, regenerated to full strength if it has been weakened, and retransmitted as light.
- If an active repeater fails, the ring is broken and the network goes down.

58. Mention the various types of wireless transmission?

The various wireless transmission are

1. Electromagnetic Spectrum
2. Radio Transmission
3. Microwave Transmission
4. Infrared Transmission

5. Light Transmission

59. What are the standards used in wireless communication? **(NOV 2015)**

- ITU – International Telecommunication Union
- IEEE – Institute of Electrical and Electronics Engineers
- IMT – International Mobile Telecommunication
- GSM – Global System for Mobile communication

60. Mention the types of spread spectrum?

1. Frequency hopping spread spectrum
2. Direct sequence spread spectrum
3. UWB (Ultra-Wide Band)

61. What is meant by IrDA?

The Infrared Data Association (IrDA), an association for sponsoring the use of infrared waves, has established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers. For example, some manufacturers provide a special port called the IrDA port that allows a wireless keyboard to communicate with a PC.

62. What are the various communication satellites?

The various communication satellites are

1. Geostationary Satellites (GEO)
2. Medium-Earth Orbit Satellites (MEO)
3. Low-Earth Orbit Satellites (LEO)

63. What are the various Frequency Bands for Satellite Communication?

Band	Downlink	Uplink	Bandwidth	Problems
L	1.5 GHz	1.6 GHz	15 MHz	Low bandwidth; crowded
S	1.9 GHz	2.2 GHz	70 MHz	Low bandwidth; crowded
C	4.0 GHz	6.0 GHz	500 MHz	Terrestrial interference
Ku	11 GHz	14 GHz	500 MHz	Rain
Ka	20 GHz	30 GHz	3500 MHz	Rain, equipment cost

11 MARKS

1. Explain in detail about the uses of Computer Networks? Explain with respective of real world applications? (11 MARKS)

Computer network'' is a collection of autonomous computers interconnected by a single technology. A Computer Network or data network is a telecommunications network which allows computers to exchange data. In computer networks, networked computing devices pass data to each other along network links (data connections). Data is transferred in the form of packets.

The various uses of Computer Networks are

1. Business Applications
2. Home Applications
3. Mobile Users
4. Social Issues

1. Business Applications

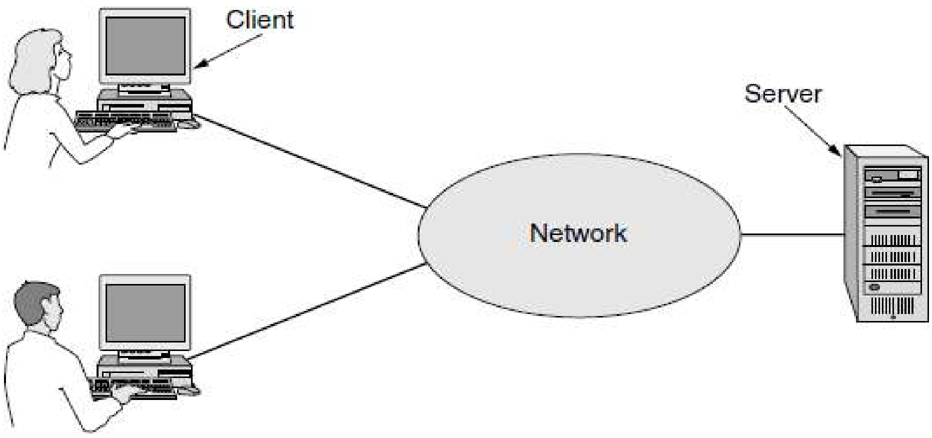
Most companies have a substantial number of computers. For example, a company may have a computer for each worker and use them to design products, write brochures, and do the payroll. Initially, some of these computers may have worked in isolation from the others, but at some point, management may have decided to connect them to be able to distribute information throughout the company.

The issues are **resource sharing**. The goal is to make all programs, equipment, and especially data available to anyone on the network without regard to the physical location of the resource or the user. An obvious and widespread example is having a group of office workers share a common printer. None of the individuals really needs a private printer, and a high-volume networked printer is often cheaper, faster, and easier to maintain than a large collection of individual printers.

However, probably even more important than sharing physical resources such as printers, and tape backup systems, is sharing information. Companies small and large are vitally dependent on computerized information. Most companies have customer records, product information, inventories, financial statements, tax information, and much more online. If all of its computers suddenly went down, a bank could not last more than five minutes. A modern manufacturing plant, with a computer-controlled assembly line, would not last even 5 seconds. Even a small travel agency or three-person law firm is now highly dependent on computer networks for allowing employees to access relevant information and documents instantly.

- Networks called **VPNs (Virtual Private Networks)** may be used to join the individual networks at different sites into one extended network. In other words, the mere fact that a user happens to be 15,000 km away from his data should not prevent him from using the data as though they were local.

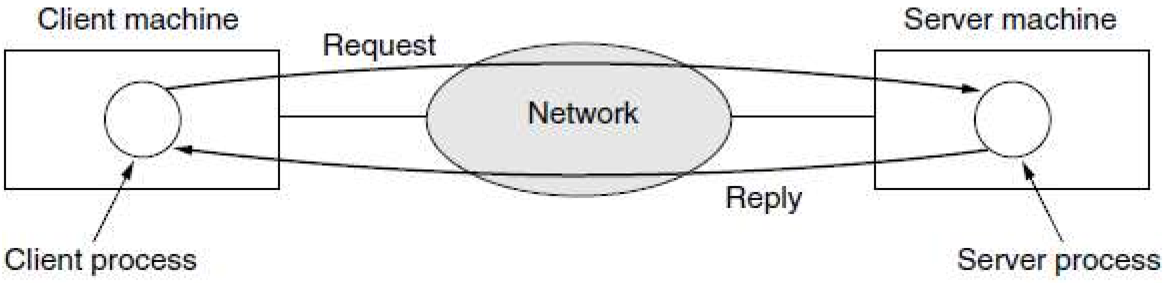
- The data are stored on powerful computers called **servers**. Often these are centrally housed and maintained by a system administrator. In contrast, the employees have simpler machines, called clients, on their desks, with which they access remote data, for example, to include in spreadsheets they are constructing. The client and server machines are connected by a network as illustrated in Figure.



A network with two clients and one server

- This whole arrangement is called the **client-server model**. It is widely used and forms the basis of much network usage. The most popular realization is that of a **Web application**, in which the server generates Web pages based on its database in response to client requests that may update the database.
- The client-server model is applicable when the client and server are both in the same building (and belong to the same company), but also when they are far apart. For example, when a person at home accesses a page on the World Wide Web, the same model is employed, with the remote Web server being the server and the user's personal computer being the client. Under most conditions, one server can handle a large number (hundreds or thousands) of clients simultaneously.

The client-server model in detail, we see that two processes (i.e., running programs) are involved, one on the client machine and one on the server machine. Communication takes the form of the client process sending a message over the network to the server process. The client process then waits for a reply message. When the server process gets the request, it performs the requested work or looks up the requested data and sends back a reply.



The client-server model involves requests and replies.

A second goal of setting up a computer network has to do with people rather than information or even computers. A computer network can provide a powerful **communication medium** among employees. Virtually every company that has two or more computers now has **email (electronic mail)**, which employees generally use for a great deal of daily communication. In fact, a common gripe around the water cooler is how much email everyone has to deal with, much of it quite meaningless because bosses have discovered that they can send the same (often content-free) message to all their subordinates at the push of a button.

Telephone calls between employees may be carried by the computer network instead of by the phone company. This technology is called IP telephony or Voice over IP (VoIP) when Internet technology is used. The microphone and speaker at each end may belong to a VoIP-enabled phone or the employee's computer. Companies find this a wonderful way to save on their telephone bills. **Desktop sharing** lets remote workers see and interact with a graphical computer screen. This makes it easy for two or more people who work far apart to read and write a shared blackboard or write a report together.

A third goal for many companies is doing business electronically, especially with customers and suppliers. This new model is called **e-commerce (electronic commerce)** and it has grown rapidly in recent years. Airlines, bookstores, and other retailers have discovered that many customers like the convenience of shop-ping from home.

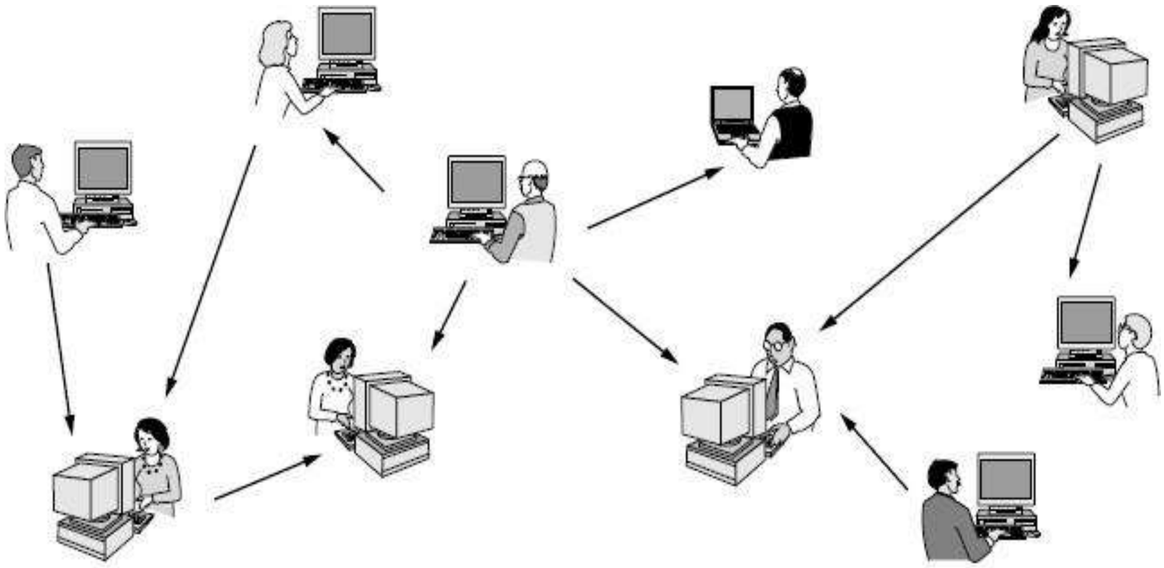
2. Home Applications

The biggest reason to buy a home computer was probably for Internet access. Now, many consumer electronic devices, such as set-top boxes, game consoles, and clock radios, come with embedded computers and computer networks, especially wireless networks, and home networks are broadly used for entertainment, including listening to, looking at, and creating music, photos, and videos.

Internet access provides home users with connectivity to remote computers. As with companies, home users can access information, communicate with other people, and buy products and services with e-commerce. The main benefit now comes from connecting outside of the home.

Access to remote information comes in many forms. It can be surfing the World Wide Web for information. Information available includes the arts, business, cooking, government, health, history, hobbies, recreation, science, sports, travel, and many others.

Much of this information is accessed using the client-server model, but there is different, popular model for accessing information that goes by the name of **peer-to-peer** communication. In this form, individuals who form a loose group can communicate with others in the group, as shown. Every person can, in principle, communicate with one or more other people; there is no fixed division into clients and servers.



In a peer-to-peer system there are no fixed clients and servers.

- **Peer-to-peer** communication is often used to share music and videos.
- There are multi-person messaging services too, such as the **Twitter** service that lets people send short text messages called ''tweets'' to their circle of friends or other willing audiences.
- The Internet can be used by applications to carry audio (e.g., Internet radio stations) and video (e.g., YouTube).
- Between person-to-person communications and accessing information are **social network** applications. Here, the flow of information is driven by the relationships that people declare between each other.
- One of the most popular social networking sites is **Facebook**. It lets people update their personal profiles and shares the updates with other people who they have declared to be their friends. Other social networking applications can make introductions via friends of friends, send news messages to friends such as Twitter above, and much more.
- Even more loosely, groups of people can work together to create content. A **wiki**, for example, is a collaborative Web site that the members of a community edit. The most famous wiki is the **Wikipedia**, an encyclopedia anyone can edit, but there are thousands of other wikis.

Another area in which **e-commerce** is widely used is access to financial institutions. Many people already pay their bills, manage their bank accounts, and handle their investments electronically. This trend will surely continue as networks become more secure.

The client-server model, online auctions are peer-to-peer in the sense that consumers can act as both buyers and sellers.

Tag	Full name	Example
B2C	Business-to-consumer	Ordering books online
B2B	Business-to-business	Car manufacturer ordering tires from supplier
G2C	Government-to-consumer	Government distributing tax forms electronically
C2C	Consumer-to-consumer	Auctioning second-hand products online
P2P	Peer-to-peer	Music sharing

Forms of E-Commerce

Our fourth category is **entertainment**. The distribution of music, radio and television programs, and movies over the Internet beginning to rival that of traditional mechanisms. Users can find, buy, and download MP3 songs and DVD-quality movies and add them to their personal collection. TV shows now reach many homes via **IPTV (IP TeleVision)** systems that are based on IP technology instead of cable TV or radio transmissions. Media streaming applications let users tune into Internet radio stations or watch recent episodes of their favorite TV shows. Naturally, all of this content can be moved around your house between different devices, displays and speakers, usually with a wireless network.

Our last category is **ubiquitous computing**, in which computing is embedded into every day. Many homes are already wired with security systems that include door and window sensors, and there are many more sensors that can be folded in to a smart home monitor, such as energy consumption. Your electricity, gas and water meters could also report usage over the network. This would save money as there would be no need to send out meter readers.

A technology called **RFID (Radio Frequency IDentification)**. RFID tags are passive (i.e., have no battery) chips the size of stamps and they can already be affixed to books, passports, pets, credit cards, and other items in the home and out. RFID readers locate and Communicate with the items over a distance of up to several meters, depending on the kind of RFID. Originally, RFID was commercialized to replace barcodes. It has not succeeded yet because barcodes are free and RFID tags cost a few cents. Of course, RFID tags offer much more and their price is rapidly declining. They may turn the real world into the Internet of things.

3. Mobile Users

Mobile computers, such as laptop and handheld computers, are one of the fastest-growing segments of the computer industry. People on the go often want to use their mobile devices to read and send email, tweet, watch movies, download music, play games, or simply to surf the Web for information. They want to do all of the things they do at home and in the office. Naturally, they want to do them from anywhere on land, sea or in the air.

Connectivity to the Internet enables many of these mobile uses. Since having a wired connection is impossible in cars, boats, and airplanes, there is a lot of interest in wireless networks. Cellular networks operated by the telephone companies are one familiar kind of wireless network that blankets us with coverage for mobile phones.

Wireless hotspots based on the 802.11 standard are another kind of wireless network for mobile computers. They have sprung up everywhere that people go, resulting in a patchwork of coverage at cafes, hotels, airports, schools, trains and planes. Anyone with a laptop computer and a wireless modem can just turn on their computer on and be connected to the Internet through the hotspot, as though the computer were plugged into a wired network.

The distinction between **fixed wireless** and **mobile wireless** networks

Wireless	Mobile	Typical applications
No	No	Desktop computers in offices
No	Yes	A notebook computer used in a hotel room
Yes	No	Networks in unwired buildings
Yes	Yes	Store inventory with a handheld computer

Combinations of wireless networks and mobile computing.

- **Text messaging or texting** is tremendously popular. It lets a mobile phone user type a short message that is then delivered by the cellular network to another mobile subscriber.
- **Smart phones**, such as the popular iPhone, combine aspects of mobile phones and mobile computers. The (3G and 4G) cellular networks to which they connect can provide fast data services for using the Internet as well as handling phone calls.
- Since mobile phones know their locations, often because they are equipped with **GPS (Global Positioning System)** receivers, some services are intentionally location dependent.
- An area in which mobile phones are now starting to be used is **m-commerce (mobile-commerce)**. Short text messages from the mobile are used to authorize payments for food in vending machines, movie tickets, and other small items instead of cash and credit cards. The charge then appears on the mobile phone bill. When equipped with **NFC (Near Field Communication)** technology the mobile can act as an RFID smartcard and interact with a nearby reader for payment.

- **Sensor networks** are made up of nodes that gather and wirelessly relay information they sense about the state of the physical world. The nodes may be part of familiar items such as cars or phones, or they may be small separate devices. For example, your car might gather data on its location, speed, vibration, and fuel efficiency from its on-board diagnostic system and upload this information to a database.
- **Wearable computers** are another promising application. Smart watches with radios have been part of our mental space. Other such devices may be implanted, such as pacemakers and insulin pumps. Some of these can be controlled over a wireless network. This lets doctors test and reconfigure them more easily. It could also lead to some nasty problems if the devices are as insecure as the average PC and can be hacked easily.

4. Social Issues

- The argument for communications that are not differentiated by their content or source or who is providing the content is known as **network neutrality.**
- For instance, pirated music and movies fueled the massive growth of peer-to-peer networks, which did not please the copyright holders, who have threatened (and sometimes taken) legal action. There are now automated systems that search peer-to-peer networks and fire off warnings to network operators and users who are suspected of infringing copyright. In the United States, these warnings are known as **DMCA takedown notices** after the **Digital Millennium Copyright Act.**
- The government does not have a monopoly on threatening people's privacy. The private sector does its bit too by **profiling** users. For example, small files called **cookies** that Web browsers store on users' computers allow companies to track users' activities in cyberspace and may also allow credit card numbers, social security numbers, and other confidential information to leak all over the Internet.
- Companies that provide Web-based services may maintain large amounts of personal information about their users that allows them to study user activities directly. For example, Google can read your email and show you advertisements based on your interests if you use its email service, **Gmail.**
- Computer networks also offer the potential to increase privacy by sending anonymous messages.
- Still other content is intended for criminal behavior. Web pages and email messages containing active content (basically, programs or macros that execute on the receiver's machine) can contain viruses that take over your computer. They might be used to steal your bank account passwords, or to have your computer send spam as part of a **botnet** or pool of compromised machines.
- **Phishing** messages masquerade as originating from a trustworthy party, for example, your bank, to try to trick you into revealing sensitive information, for example, credit card numbers.

2. Explain in detail about Network Hardware and types of Networks? (11 MARKS)

There are two types of transmission technology that are in widespread use:

1. broadcast links
2. point-to-point links

- **Point-to-point links** connect individual pairs of machines. To go from the source to the destination on a network made up of point-to-point links, short messages, called **packets** in certain contexts, may have to first visit one or more intermediate machines. Often multiple routes, of different lengths, are possible, so finding good ones is important in point-to-point networks. Point-to-point transmission with exactly one sender and exactly one receiver is sometimes called **unicasting.**
- In contrast, on a **broadcast network**, the communication channel is shared by all the machines on the network; packets sent by any machine are received by all the others. An address field within each packet specifies the intended recipient. Upon receiving a packet, a machine checks the address field. If the packet is intended for the receiving machine, that machine processes the packet; if the packet is intended for some other machine, it is just ignored.
- A **wireless network** is a common example of a broadcast link, with communication shared over a coverage region that depends on the wireless channel and the transmitting machine.
- Broadcast systems usually also allow the possibility of addressing a packet to all destinations by using a special code in the address field. When a packet with this code is transmitted, it is received and processed by every machine on the network. This mode of operation is called **broadcasting.** Some broadcast systems also support transmission to a subset of the machines, which known as **multicasting.**
- We classify multiple processor systems by their rough physical size. At the top are the personal area networks, networks that are meant for one person. Beyond these come longer-range networks. These can be divided into local, metropolitan, and wide area networks, each with increasing scale. Finally, the connection of two or more networks is called an **internetwork**. The worldwide Internet is certainly the best-known example of an internetwork.

Interprocessor distance	Processors located in same	Example
1 m	Square meter	Personal area network
10 m	Room	
100 m	Building	Local area network
1 km	Campus	
10 km	City	Metropolitan area network
100 km	Country	Wide area network
1000 km	Continent	
10,000 km	Planet	The Internet

Classification of interconnected processors by scale

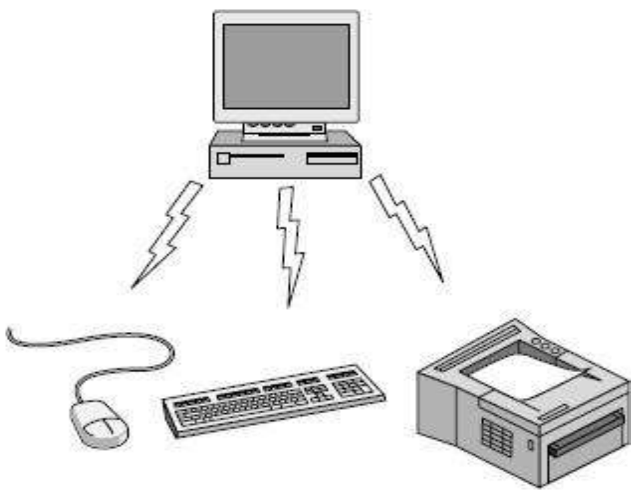
The various types of networks are

1. Personal Area Networks (PAN)
2. Local Area Networks (LAN)
3. Metropolitan Area Networks (MAN)
4. Wide Area Networks (WAN)
5. Internetworks

1. Personal Area Networks (PAN)

- PANs (Personal Area Networks) let devices communicate over the range of a person. A common example is a wireless network that connects a computer with its peripherals. Almost every computer has an attached monitor, keyboard, mouse, and printer.
- Without using wireless, this connection must be done with cables. So many new users have a hard time finding the right cables and plugging them into the right little holes (even though they are usually color coded) that most computer vendors offer the option of sending a technician to the user's home to do it.
- To help these users, some companies got together to design a short-range wireless network called **Bluetooth** to connect these components without wires. The idea is that if your devices have Bluetooth, then you need no cables. You just put them down, turn them on, and they work together.

In the simplest form, **Bluetooth** networks use the master-slave paradigm of Figure. The system unit (the PC) is normally the master, talking to the mouse, keyboard, etc., as slaves. The master tells the slaves what addresses to use, when they can broadcast, how long they can transmit, what frequencies they can use, and so on.

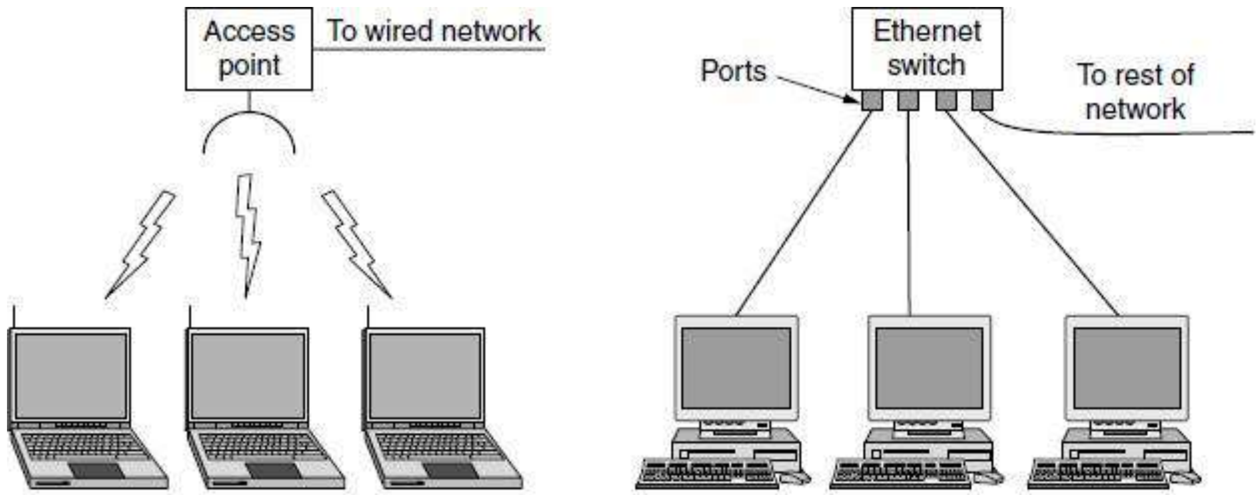


Bluetooth PAN Configuration

Bluetooth can be used in other settings, too. It is often used to connect a headset to a mobile phone without cords and it can allow your digital music player to connect to your car merely being brought within range. A completely different kind of PAN is formed when an embedded medical device such as a pacemaker, insulin pump, or hearing aid talks to a user-operated remote control.

2. Local Area Networks (LAN)

- A LAN (Local Area Network) is a privately owned network that operates within and nearby a single building like a home, office or factory.
- LANs are widely used to connect personal computers and consumer electronics to let them share resources (e.g., printers) and exchange information. When LANs are used by companies, they are called **enterprise networks.**
- Wireless LANs are very popular these days, especially in homes, older office buildings, cafeterias, and other places where it is too much trouble to install cables. In these systems, every computer has a radio modem and an antenna that it uses to communicate with other computers.
- In most cases, each computer talks to a device in the ceiling as shown in Figure (a). This device, called an **AP (Access Point)**, wireless router, or base station, relays packets between the wireless computers and also between them and the Internet. However, if other computers are close enough, they can communicate directly with one another in a peer-to-peer configuration.

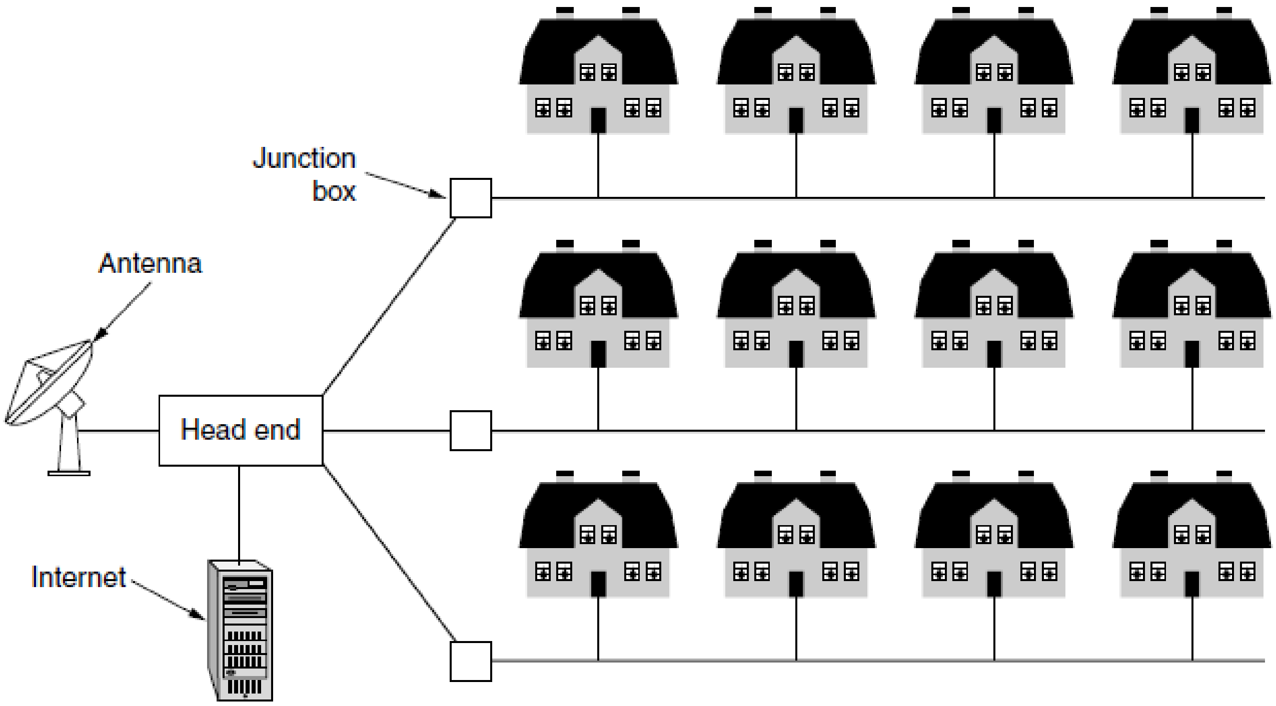


Wireless and wired LANs. (a) 802.11. (b) Switched Ethernet.

- There is a standard for wireless LANs called IEEE 802.11, popularly known as WiFi, which has become very widespread. It runs at speeds anywhere from 11 to hundreds of Mbps.
- Wired LANs use a range of different transmission technologies. Most of them use copper wires, but some use optical fiber. LANs are restricted in size, which means that the worst-case transmission time is bounded and known in advance.
- Typically, wired LANs run at speeds of 100 Mbps to 1 Gbps, have low delay (microseconds or nanoseconds), and make very few errors. Newer LANs can operate at up to 10 Gbps. Compared to wireless networks, wired LANs exceed them in all dimensions of performance.
- The topology of many wired LANs is built from point-to-point links. IEEE 802.3, popularly called **Ethernet**, is, by far, the most common type of wired LAN Figure (b) shows a sample topology of **switched Ethernet.** Each computer speaks the Ethernet protocol and connects to a box called a **switch** with a point-to-point link. A switch has multiple **ports**, each of which can connect to one computer. The job of the switch is to relay packets between computers that are attached to it, using the address in each packet to determine which computer to send it to.

3. Metropolitan Area Networks (MAN)

- MAN (Metropolitan Area Network) covers a city. The best-known examples of MANs are the cable television networks available in many cities. These systems grew from earlier community antenna systems used in areas with poor over-the-air television reception.
- In those early systems, a large antenna was placed on top of a nearby hill and a signal was then piped to the subscribers' houses.
- When the Internet began attracting a mass audience, the cable TV network operators began to realize that with some changes to the system, they could provide two-way Internet service in unused parts of the spectrum. At that point, the cable TV system began to morph from simply a way to distribute television to a metropolitan area network. To a first approximation, a MAN might look something like the system shown in Figure.
- In this figure we see both television signals and Internet being fed into the centralized cable **headend** for subsequent distribution to people's homes.

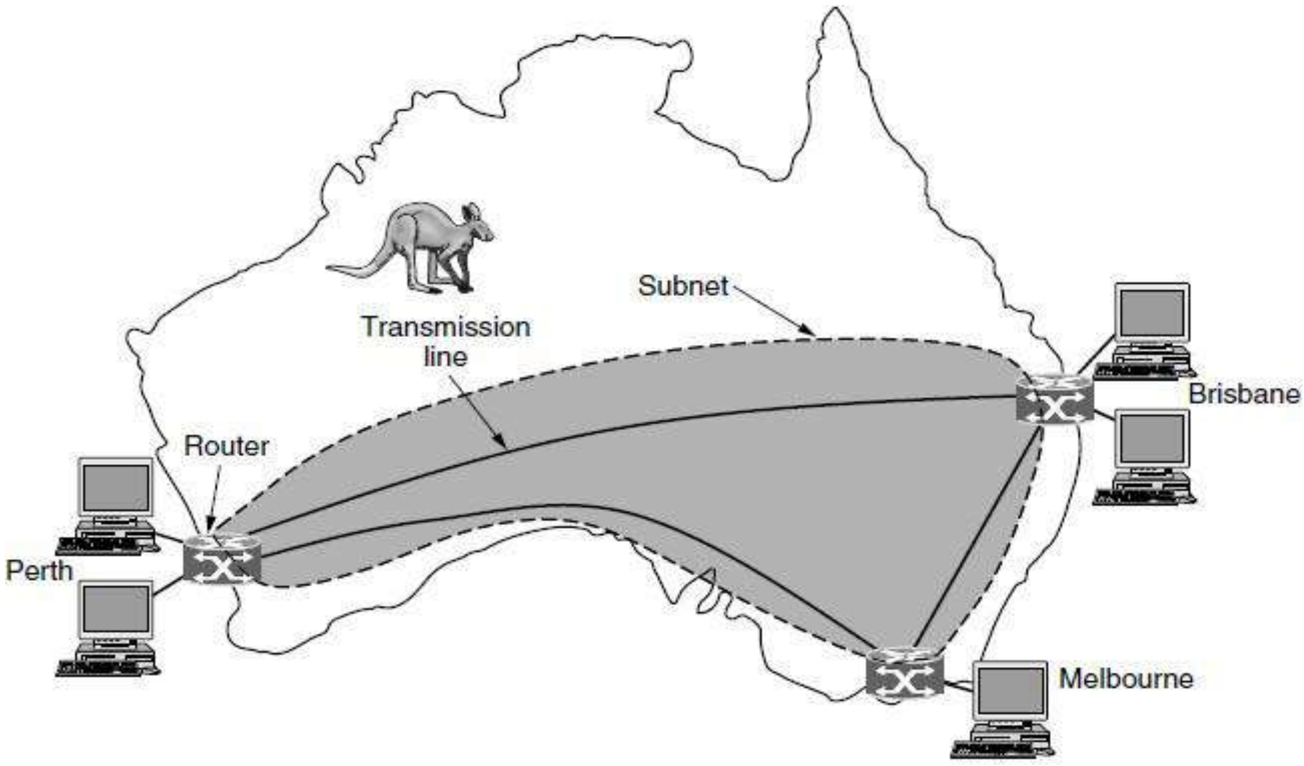


A metropolitan area network based on cable TV.

Cable television is not the only MAN, though. Recent developments in high speed wireless Internet access have resulted in another MAN, which has been standardized as IEEE 802.16 and is popularly known as **WiMAX.**

4. Wide Area Networks

- A WAN (Wide Area Network) spans a large geographical area, often a country or continent.
- The WAN in Figure is a network that connects offices in Perth, Melbourne, and Brisbane.



WAN that connects three branch offices in Australia.

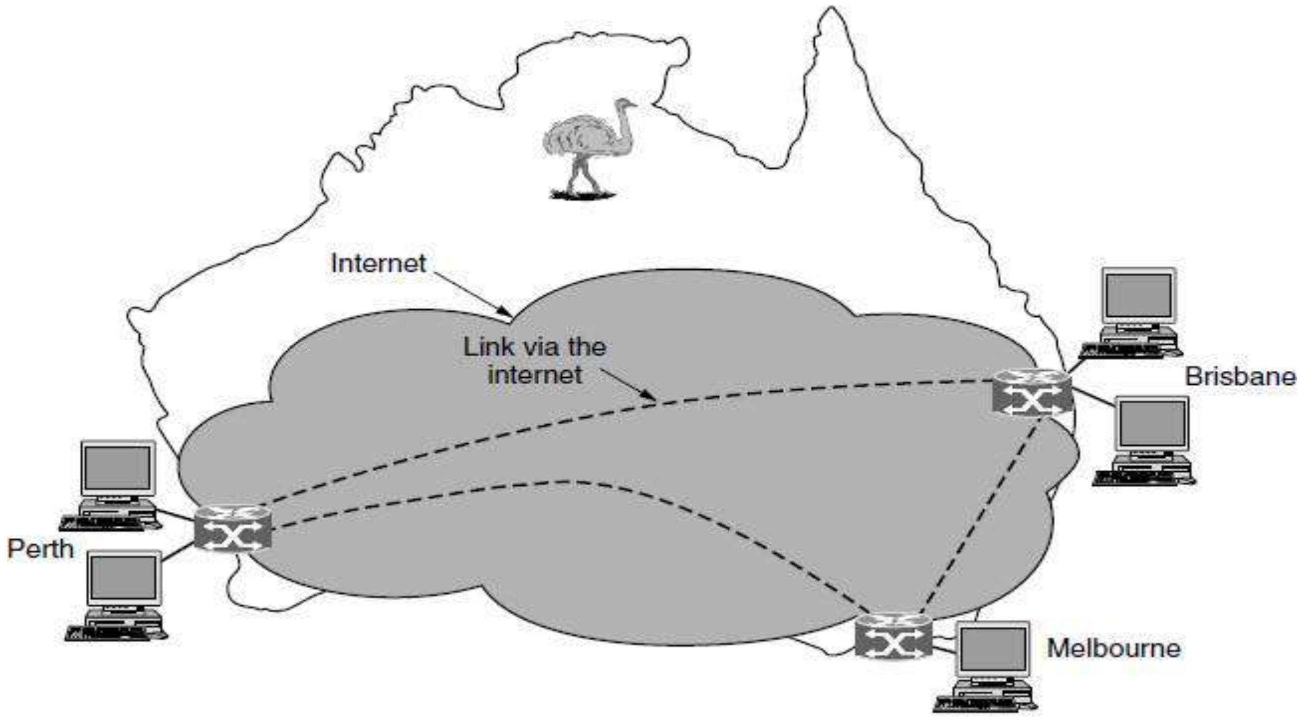
- Each of these offices contains computers intended for running user (i.e., application) programs. We will follow traditional usage and call these machines **hosts**. The rest of the network that connects these hosts is then called the **communication subnet**, or just **subnet.**

In most WANs, the subnet consists of two distinct components:

- **Transmission lines** and
- **Switching elements**

- Transmission lines move bits between machines. They can be made of copper wire, optical fiber, or even radio links. Most companies do not have transmission lines lying about, so instead they lease the lines from a telecommunications company.
- Switching elements, or just switches, are specialized computers that connect two or more transmission lines. When data arrive on an incoming line, the switching element must choose an outgoing line on which to forward them.

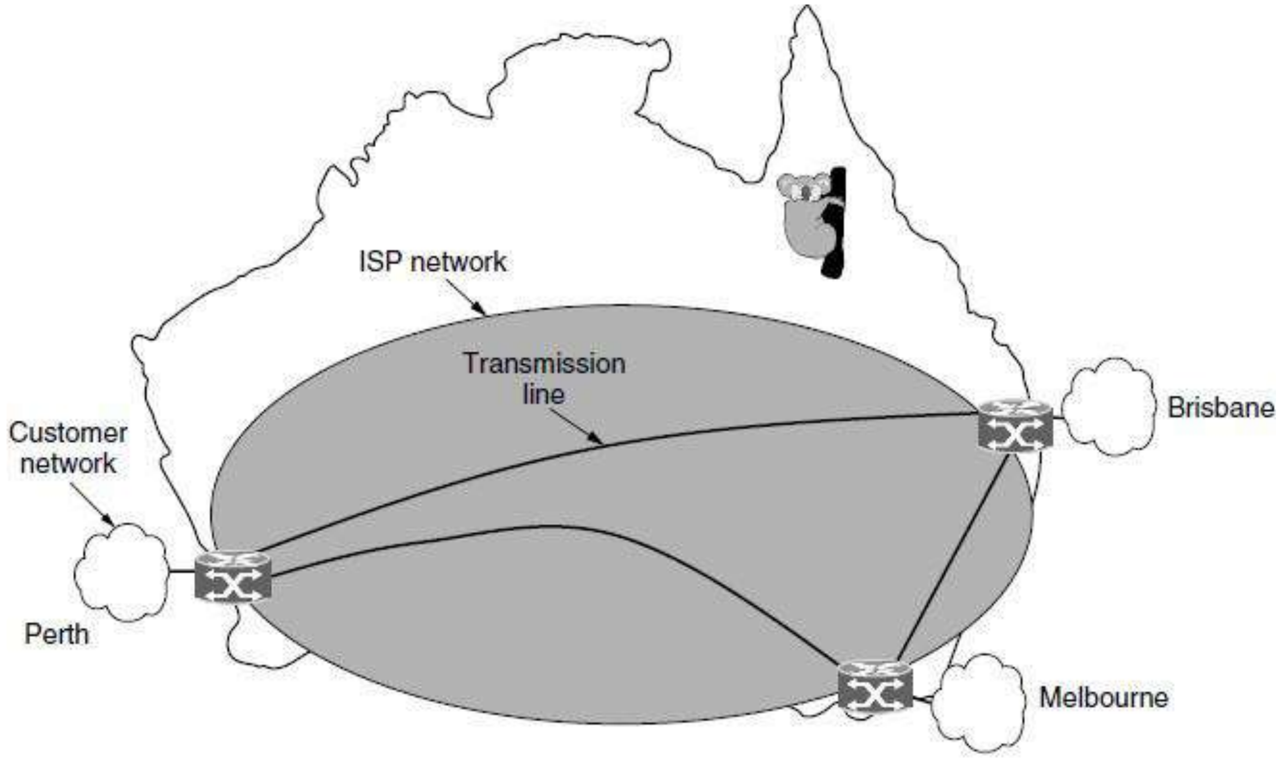
The dedicated transmission lines, a company might connect its offices to the Internet. This allows connections to be made between the offices as virtual links that use the underlying capacity of the Internet. This is called a **VPN (Virtual Private Network).**



WAN using a virtual private network.

Compared to the dedicated arrangement, a VPN has the usual advantage of virtualization, which is that it provides flexible reuse of a resource (Internet connectivity). A VPN also has the usual disadvantage of virtualization, which is a lack of control over the underlying resources.

The second variation is that the subnet may be run by a different company. The subnet operator is known as a **network service provider** and the offices are its customers. This structure is shown in Figure.



WAN using an ISP network.

The subnet operator will connect to other customers too, as long as they can pay and it can provide service. Since it would be a disappointing network service if the customers could only send packets to each other, the subnet operator will also connect to other networks that are part of the Internet. Such a subnet operator is called an **ISP (Internet Service Provider)** and the subnet is an **ISP network**. Its customers who connect to the ISP receive Internet service.

5. Internetworks

- Many networks exist in the world, often with different hardware and software. People connected to one network often want to communicate with people attached to a different one.
- The fulfillment of this desire requires that different, and frequently incompatible, networks be connected. A collection of interconnected networks is called an **internetwork** or **internet.**
- The Internet uses ISP networks to connect enterprise networks, home networks, and many other networks.
- The ‘‘subnet’’ makes the most sense in the context of a wide area network, where it refers to the collection of routers and communication lines owned by the network operator.
- A network is formed by the combination of a subnet and its hosts.
- An internet is formed when distinct networks are interconnected. In our view, connecting a LAN and a WAN or connecting two LANs is the usual way to form an internetwork.

There are two rules of thumb that are useful.

- First, if different organizations have paid to construct different parts of the network and each maintains its part, we have an internetwork rather than a single network.
- Second, if the underlying technology is different in different parts (e.g., broadcast versus point-to-point and wired versus wireless), we probably have an internetwork.

- A machine that makes a connection between two or more networks and provides the necessary translation, both in terms of hardware and software, is a **gateway**.
- Gateways are distinguished by the layer at which they operate in the protocol hierarchy.

3. Explain in detail about Protocol Hierarchy in Network Software? (6 MARKS)

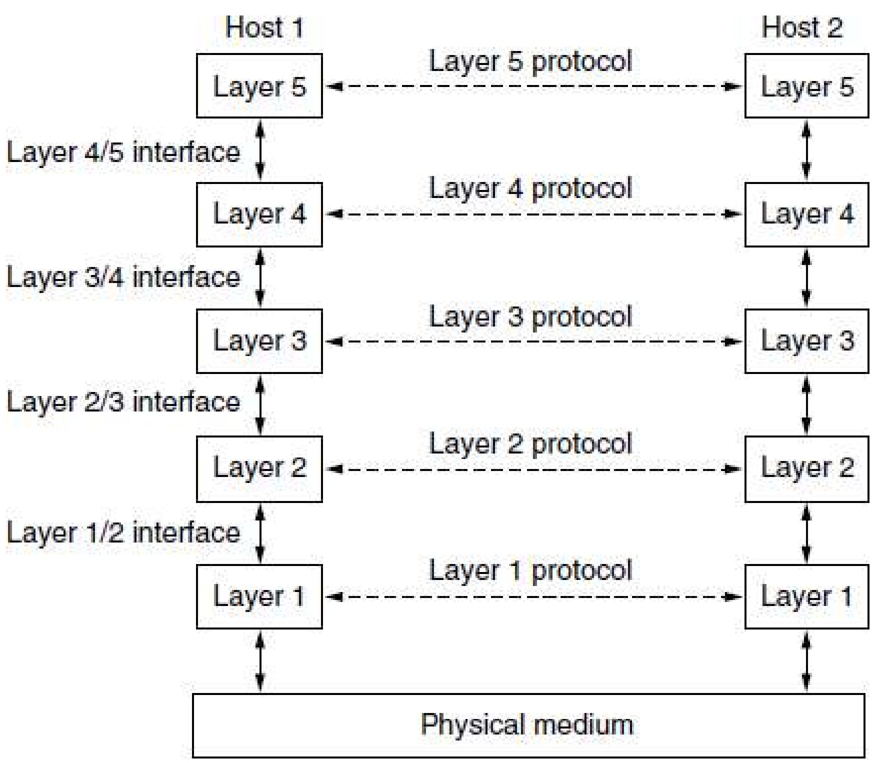
Protocol Hierarchy:

- To reduce their design complexity, most networks are organized as a stack of layers or levels, each one built upon the one below it. The number of layers, the name of each layer, the contents of each layer, and the function of each layer differ from network to network.
- The purpose of each layer is to offer certain services to the higher layers while shielding those layers from the details of how the Offered services are actually implemented. In a sense, each layer is a kind of virtual machine, offering certain services to the layer above it.
- The fundamental idea is that a particular piece of software (or hardware) provides a service to its users but keeps the details of its internal state and algorithms hidden from them.

When layer n on one machine carries on a conversation with layer n on another machine, the rules and conventions used in this conversation are collectively known as the layer n protocol.

A **protocol** is a set of rules that govern data communications. A protocol defines what is communicated, how it is communicated, and when it is communicated. The key elements of a protocol are syntax, semantics, and timing.

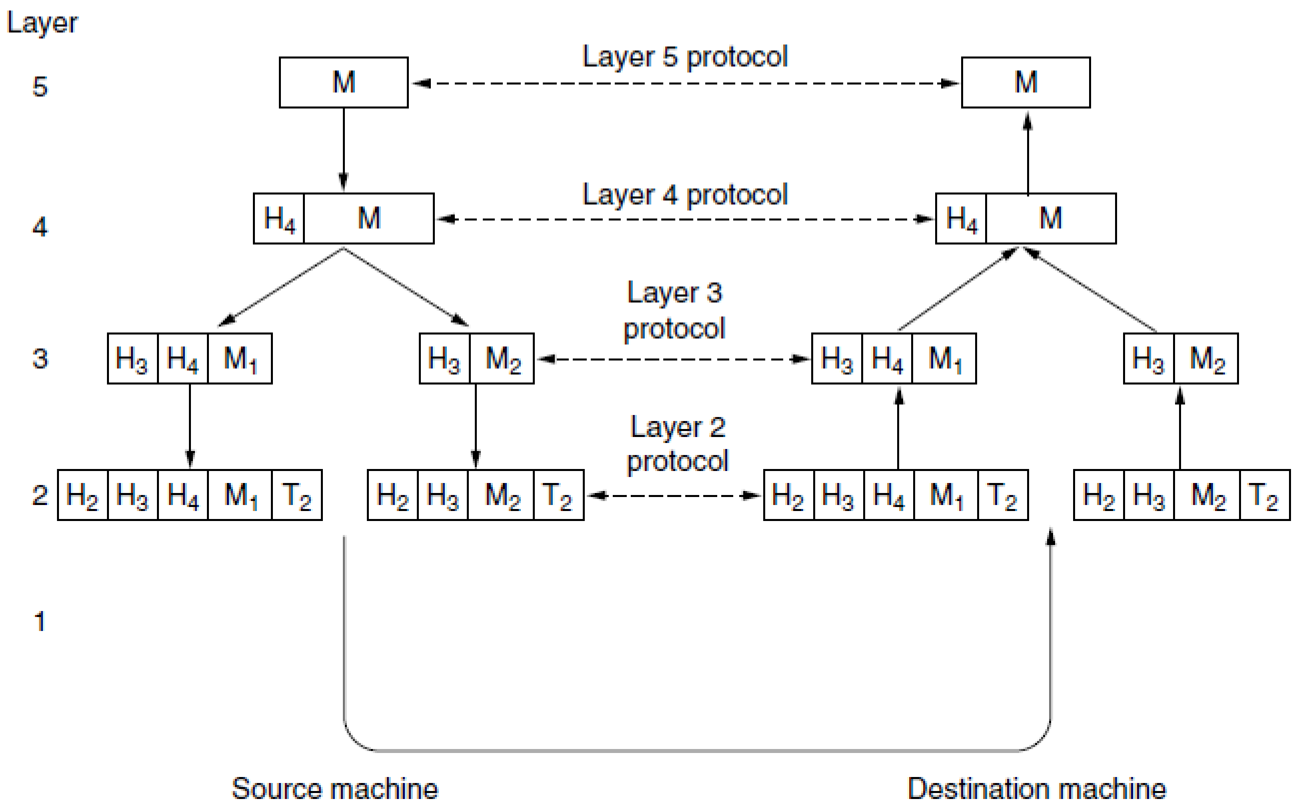
A five-layer network is illustrated in Figure. The entities comprising the corresponding layers on different machines are called **peers.** The peers may be software processes, hardware devices, or even human beings. In other words, it is the peers that communicate by using the protocol to talk to each other.



Layers, protocols, and interfaces.

- In reality, no data are directly transferred from layer n on one machine to layer n on another machine. Instead, each layer passes data and control information to the layer immediately below it, until the lowest layer is reached. Below layer 1 is the physical medium through which actual communication occurs.
- In Figure, virtual communication is shown by dotted lines and physical communication by solid lines.
- Between each pair of adjacent layers is an **interface.** The interface defines which primitive operations and services the lower layer makes available to the upper one. When network designers decide how many layers to include in a network and what each one should do, one of the most important considerations is defining clean interfaces between the layers.
- A set of layers and protocols is called a **network architecture.** The specification of an architecture must contain enough information to allow an implementer to write the program or build the hardware for each layer so that it will correctly obey the appropriate protocol.
- A list of the protocols used by a certain system, one protocol per layer, is called a **protocol stack.**

Now consider a more technical example: how to provide communication to the top layer of the five-layer network in Figure. A message, M, is produced by an application process running in layer 5 and given to layer 4 for transmission. Layer 4 puts a header in front of the message to identify the message and passes the result to layer 3. The header includes control information, such as addresses, to allow layer 4 on the destination machine to deliver the message. Other examples of control information used in some layers are sequence number, sizes, and times.



Example information flow supporting virtual communication in layer 5

4. Explain the design issues for layers in Network Software? (5 MARKS)

Some of the key design issues that occur in computer networks are

Reliability is the design issue of making a network that operates correctly even though it is made up of a collection of components that are themselves unreliable. Think about the bits of a packet traveling through the network.

There is a chance that some of these bits will be received damaged (inverted) due to fluke electrical noise, random wireless signals, hardware flaws, software bugs and so on. How is it possible that we find and fix these errors?

1. One mechanism for finding errors in received information uses codes for **error detection**. Information that is incorrectly received can then be retransmitted until it is received correctly. More powerful codes allow for error correction, where the correct message is recovered from the possibly incorrect bits that were originally received. Both of these mechanisms work by adding redundant information. They are used at low layers, to protect packets sent over individual links, and high layers, to check that the right contents were received.
   - Another reliability issue is finding a working path through a network. Often there are multiple paths between a source and destination, and in a large network, here may be some links or routers that are broken.
2. A second design issue concerns the evolution of the network. Over time, networks grow larger and new designs emerge that need to be connected to the existing network. The key structuring mechanism used to support change by dividing the overall problem and hiding implementation details: **protocol layering.**
   - Since there are many computers on the network, every layer needs a mechanism for identifying the senders and receivers that are involved in a particular message. This mechanism is called **addressing or naming**, in the low and high layers, respectively.
   - When networks get large, new problems arise. Cities can have traffic jams, a shortage of telephone numbers, and it is easy to get lost. Designs that continue to work well when the network gets large are said to be **scalable.**

3. A third design issue is resource allocation. Networks provide a service to hosts from their underlying resources, such as the capacity of transmission lines. They need mechanisms that divide their resources so that one host does not interfere with another too much.
   - Many designs share network bandwidth dynamically, according to the short term needs of hosts, rather than by giving each host a fixed fraction of the bandwidth that it may or may not use. This design is called statistical multiplexing, meaning sharing based on the statistics of demand.

An allocation problem that occurs at every level is how to keep a fast sender from swamping a slow receiver with data. Feedback from the receiver to the sender is often used. This subject is called **flow control.**

Sometimes the problem is that the network is oversubscribed because too many computers want to send too much traffic, and the network cannot deliver it all. This overloading of the network is called **congestion**.

It is interesting to observe that the network has more resources to offer than simply bandwidth. For uses such as carrying live video, the timeliness of delivery matters a great deal. Most networks must provide service to applications that want this real-time delivery at the same time that they provide service to applications that want high throughput. **Quality of service** is the name given to mechanisms that reconcile these competing demands.

The last major design issue is to secure the network by defending it against different kinds of threats. One of the threats we have mentioned previously is that of eavesdropping on communications.

- Mechanisms that provide **confidentiality** defend against this threat, and they are used in multiple layers.
- Mechanisms for **authentication** prevent someone from impersonating someone else. They might be used to tell fake banking Web sites from the real one, or to let the cellular network check that a call is really coming from your phone so that you will pay the bill.
- Other mechanisms for **integrity** prevent surreptitious changes to messages, such as altering "debit my account $10" to "debit my account $1000."

5. Compare the Connection-Oriented Versus Connectionless Service? (6 MARKS)

Layers can offer two different types of service to the layers are

1. Connection-oriented Service and
2. Connection-less Service

Connection-oriented Service:

- Connection-oriented service is modeled after the telephone system. To talk to someone, you pick up the phone, dial the number, talk, and then hang up.
- Similarly, to use a connection-oriented network service, the service user first establishes a connection, uses the connection, and then releases the connection.
- The essential aspect of a connection is that it acts like a tube: the sender pushes objects (bits) in at one end, and the receiver takes them out at the other end. In most cases the order is preserved so that the bits arrive in the order they were sent.
- In some cases when a connection is established, the sender, receiver, and subnet conduct a negotiation about the parameters to be used, such as maximum message size, quality of service required, and other issues. Typically, one side makes a proposal and the other side can accept it, reject it.
- A circuit is another name for a connection with associated resources, such as a fixed bandwidth. This dates from the telephone network in which a circuit was a path over copper wire that carried a phone conversation.

Connection-less Service:

- In contrast to connection-oriented service, connectionless service is modeled after the postal system.
- Each message (letter) carries the full destination address, and each one is routed through the intermediate nodes inside the system independent of all the subsequent messages.
- There are different names for messages in different contexts; a **packet** is a message at the network layer.
- When the intermediate nodes receive a message in full before sending it on to the next node, this is called **store-and-forward switching**.
- The alternative, in which the onward transmission of a message at a node starts before it is completely received by the node, is called **cut-through switching.**
- Normally, when two messages are sent to the same destination, the first one sent will be the first one to arrive. However, it is possible that the first one sent can be delayed so that the second one arrives first.

Each kind of service can further be characterized by its reliability. Some services are reliable in the sense that they never lose data. Usually, a reliable service is implemented by having the receiver acknowledge the receipt of each message so the sender is sure that it arrived. The acknowledgement process introduces overhead and delays, which are often worth it but are sometimes undesirable.

A typical situation in which a reliable connection-oriented service is appropriate is file transfer. The owner of the file wants to be sure that all the bits arrive correctly and in the same order they were sent. Very few file transfer customers would prefer a service that occasionally scrambles or loses a few bits, even if it is much faster.

Reliable connection-oriented service has two minor variations: **message sequences** and **byte streams.**

For some applications, the transit delays introduced by acknowledgements are unacceptable. One such application is digitized voice traffic for **voice over IP**. It is less disruptive for telephone users to hear a bit of noise on the line from time to time than to experience a delay waiting for acknowledgements. Similarly, when transmitting a video conference, having a few pixels wrong is no problem, but having the image jerk along as the flow stops and starts to correct errors is irritating.

- Unreliable (meaning not acknowledged) connectionless service is often called **datagram service**, in analogy with telegram service, which also does not return an acknowledgement to the sender.
- In other situations, the convenience of not having to establish a connection to send one message is desired, but reliability is essential. The **acknowledged datagram service** can be provided for these applications. It is like sending a registered letter and requesting a return receipt. When the receipt comes back, the sender is absolutely sure that the letter was delivered to the intended party and not lost along the way. Text messaging on mobile phones is an example.

- Still another service is the **request-reply service**. In this service the sender transmits a single datagram containing a request; the reply contains the answer. Request-reply is commonly used to implement communication in the client-server model: the client issues a request and the server responds to it. For example, a mobile phone client might send a query to a map server to retrieve the map data for the current location.

Types of services

	Service	Example
Connection-oriented	Reliable message stream	Sequence of pages
	Reliable byte stream	Movie download
	Unreliable connection	Voice over IP
Connection-less	Unreliable datagram	Electronic junk mail
	Acknowledged datagram	Text messaging
	Request-reply	Database query

6. Write a short on Socket Primitives (OR) Service Primitives (5 MARKS)

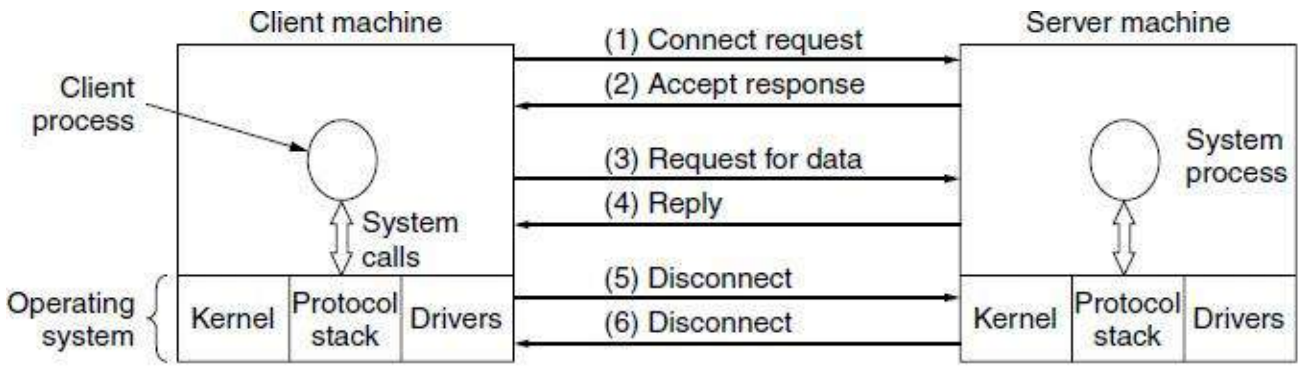
- A service is formally specified by a set of **primitives** (operations) available to user processes to access the service. These primitives tell the service to perform some action or report on an action taken by a peer entity.
- If the protocol stack is located in the operating system, as it often is, the primitives are normally system calls. These calls cause a trap to kernel mode, which then turns control of the machine over to the operating system to send the necessary packets.
- The set of primitives available depends on the nature of the service being provided. The primitives for connection-oriented service are different from those of connectionless service.

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
ACCEPT	Accept an incoming connection from a peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

Six service primitives that provide a simple connection-oriented service

These primitives might be used for a request-reply interaction in a client-server environment.

1. First, the server executes LISTEN to indicate that it is prepared to accept incoming connections. A common way to implement LISTEN is to make it a blocking system call. After executing the primitive, the server process is blocked until a request for connection appears.
2. Next, the client process executes CONNECT to establish a connection with the server. The CONNECT call needs to specify who to connect to, so it might have a parameter giving the server's address. The operating system then typically sends a packet to the peer asking it to connect, as shown by (1) in Figure. The client process is suspended until there is a response.

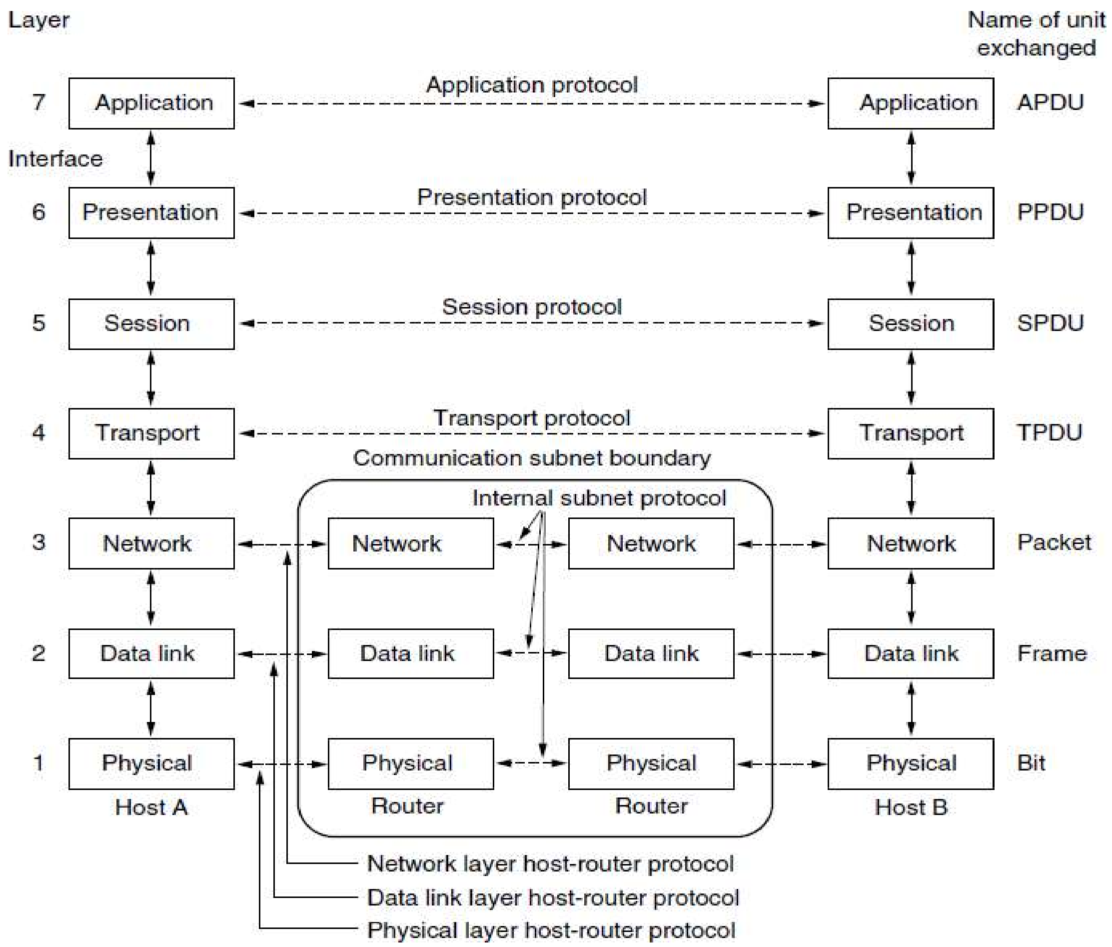


A simple client-server interaction using acknowledged datagrams.

3. When the packet arrives at the server, the operating system sees that the packet is requesting a connection. It checks to see if there is a listener, and if so it unblocks the listener. The server process can then establish the connection with the ACCEPT call. This sends a response (2) back to the client process to accept the connection. The arrival of this response then releases the client. At this point the client and server are both running and they have a connection established.
4. The next step is for the server to execute RECEIVE to prepare to accept the first request. Normally, the server does this immediately upon being released from the LISTEN, before the acknowledgement can get back to the client. The RECEIVE call blocks the server.
5. Then the client executes SEND to transmit its request (3) followed by the execution of RECEIVE to get the reply. The arrival of the request packet at the server machine unblocks the server so it can handle the request. After it has done the work, the server uses SEND to return the answer to the client (4). The arrival of this packet unblocks the client, which can now inspect the answer. If the client has additional requests, it can make them now.
6. When the client is done, it executes DISCONNECT to terminate the connection (5). Usually, an initial DISCONNECT is a blocking call, suspending the client and sending a packet to the server saying that the connection is no longer needed. When the server gets the packet, it also issues a DISCONNECT of its own, acknowledging the client and releasing the connection (6). When the server's packet gets back to the client machine, the client process is released and the connection is broken. In a nutshell, this is how connection-oriented communication works.

7. Explain in detail about OSI Reference Model? (11 MARKS) (NOV 2015, 2016) (MAY 2016)

- The model is based on a proposal developed by the **International Standards Organization (ISO)** as a first step toward international standardization of the protocols used in the various layers.
- The model is called the **ISO OSI (Open Systems Interconnection)** Reference Model, it deals with connecting open systems—that is, systems that are open for communication with other systems.
- The OSI model has seven layers.



The OSI reference model

1. Physical Layer

The physical layer coordinates the functions required to carry a bit stream over a physical medium. The design issues have to do with making sure that when one side sends a 1 bit it is received by the other side as a 1 bit, not as a 0 bit. It deals with the mechanical and electrical specifications of the interface and transmission medium. It also defines the procedures and functions that physical devices and interfaces have to perform for transmission to occur.

- **Physical characteristics of interfaces and media** - The physical layer defines the characteristics of the interface between the devices and the transmission medium.
- **Representation of bits** – The physical layer data consists of a stream of bits (sequence of 0s or 1s) with no interpretation. To be transmitted, bits must be encoded into signals--electrical or optical. The physical layer defines the type of encoding.

- **Data Rate or Transmission rate -** The number of bits sent each second – is also defined by the physical layer.

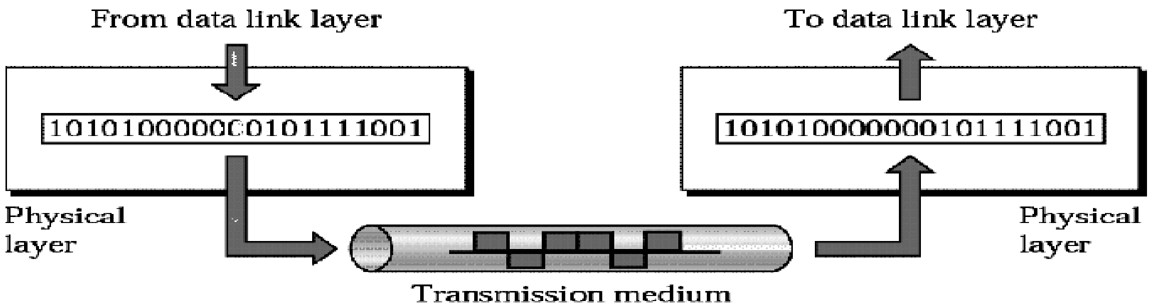


Fig. Position of the physical layer with respect to the transmission medium and the data link layer.

The physical layer is concerned with the following:

- **Synchronization of bits** - The sender and receiver must be synchronized at the bit level. Their clocks must be synchronized.
- **Line Configuration** - In a point-to-point configuration, two devices are connected together through a dedicated link. In a multipoint configuration, a link is shared between several devices.

2. Data Link Layer

The data link layer transforms the physical layer, a raw transmission facility, to a reliable link. It makes the physical layer appear error-free to the upper layer (network layer).

Other responsibilities of the data link layer includethe following:

- **Framing** - The data link layer divides the stream of bits received from the network layer into manageable data units called frames.
- **Flow control** - If the rate at which the data are absorbed by the receiver is less than the rate at which data are produced in the sender, the data link layer imposes a flow control mechanism to avoid overwhelming the receiver.
- **Error control** - The data link layer adds reliability to the physical layer by adding mechanisms to detect and retransmit damaged or lost frames. It also uses a mechanism to recognize duplicate frames. Error controls normally achieved through a trailer added to the end of the frame.

3. Network Layer

The network layer is responsible for the source-to-destination delivery of a packet, possibly across multiple networks (links). Whereas the data link layer oversees the delivery of the packet between two systems on the same network (links), the network layer ensures that each packet gets from its point of origin to its final destination.

If too many packets are present in the subnet at the same time, they will get in one another's way, forming bottlenecks. Handling congestion is also a responsibility of the network layer, in conjunction with higher layers that adapt the load they place on the network. More generally, the quality of service provided (delay, transit time, jitter, etc.) is also a network layer issue.

Other responsibilities of the network layer include the following:

- **Logical addressing** - The physical addressing implemented by the data link layer handles the addressing problem locally. If a packet passes the network boundary, we need another addressing system to help distinguish the source and destination systems. The network layer adds a header to the packet coming from the upper layer that, among other things, includes the logical addresses of the sender and receiver.
- **Routing** - When independent networks or links are connected to create internetworks (network of networks) or a large network, the connecting devices (called routers or switches) route or switch the packets to their final destination.

4. Transport Layer

The basic function of the **transport layer** is to accept data from above it, split it up into smaller units if need be, pass these to the network layer, and ensure that the pieces all arrive correctly at the other end. Furthermore, all this must be done efficiently and in a way that isolates the upper layers from the inevitable changes in the hardware technology over the course of time.

The transport layer also determines what type of service to provide to the session layer, and, ultimately, to the users of the network. The most popular type of transport connection is an error-free point-to-point channel that delivers messages or bytes in the order in which they were sent.

Other responsibilities of the transport layer include the following:

- **Flow control** - The transport layer is responsible for flow control. However, flow control at this layer is performed end to end rather than across a single link.
- **Error control** - The transport layer is responsible for error control. However, error control at this layer is performed process-to process rather than across a single link. The sending transport layer makes sure that the entire message arrives at the receiving transport layer without error (damage, loss, or duplication). Error correction is usually achieved through retransmission.

5. Session Layer

The session layer allows users on different machines to establish sessions between them. Sessions offer various services, including dialog control (keeping track of whose turn it is to transmit), token management (preventing two parties from attempting the same critical operation simultaneously), and synchronization (checkpointing long transmissions to allow them to pick up from where they left off in the event of a crash and subsequent recovery).

Specific responsibilities of the session layer include the following:

- **Dialog control** - The session layer allows two systems to enter into a dialog. It allows the communication between two processes to take place in either half-duplex (one way at a time) or full-duplex (two ways at a time) mode.

- **Synchronization** - The session layer allows a process to add checkpoints, or synchronization points, to a stream of data.

6. Presentation Layer

The presentation layer is concerned with the syntax and semantics of the information transmitted. The presentation layer is responsible for translation, compression, and encryption. In order to make it possible for computers with different internal data representations to communicate, the data structures to be exchanged can be defined in an abstract way, along with a standard encoding to be used ''on the wire.'' The presentation layer manages these abstract data structures and allows higher-level data structures (e.g., banking records) to be defined and exchanged.

7. Application Layer

- The application layer enables the user, whether human or software, to access the network. It provides user interfaces and support for services such as electronic mail, remote file access and transfer, shared database management, and other types of distributed information services.
- The application layer is responsible for providing services to the user.

The application layer contains a variety of protocols that are commonly needed by users. One widely used application protocol is HTTP (Hyper Text Transfer Protocol), which is the basis for the World Wide Web. When a browser wants a Web page, it sends the name of the page it wants to the server hosting the page using HTTP. The server then sends the page back. Other application protocols are used for file transfer, electronic mail, and network news.

8. Explain in detail about TCP/IP Reference Model? (11 MARKS)

The original TCP/IP protocol suite was defined as having four layers:

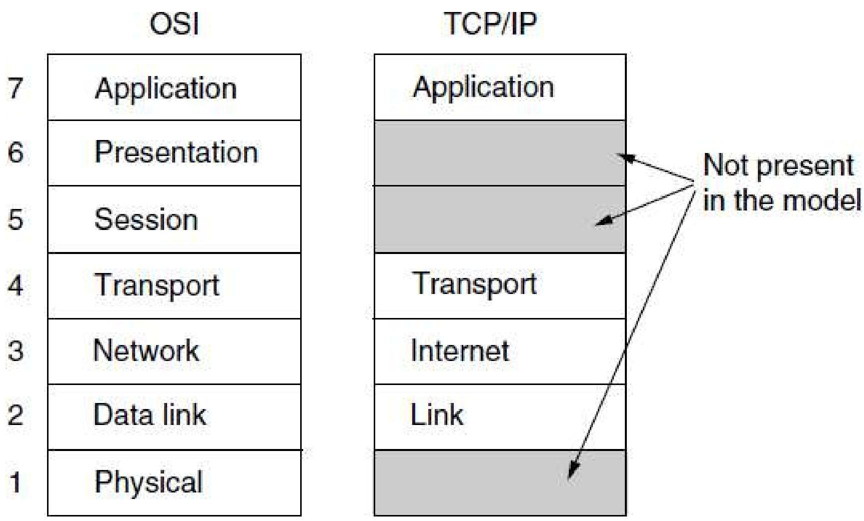
1. Link
2. Internet
3. Transport
4. Application

1. The Link Layer

A packet-switching network based on a connectionless layer that runs across different networks. The lowest layer in the model, the link layer describes what links such as serial lines and classic Ethernet must do to meet the needs of this connectionless internet layer.

2. The Internet Layer

The internet layer is the essential that holds the whole architecture together. It is shown in Figure as corresponding roughly to the OSI network layer. Its job is to permit hosts to inject packets into any network and have them travel independently to the destination (potentially on a different network). They may even arrive in a completely different order than they were sent, in which case it is the job of higher layers to rearrange them, if in-order delivery is desired.



The TCP/IP reference model

The analogy here is with the (e-mail) mail system. A person can drop a sequence of international letters into a mailbox in one country, and most of them will be delivered to the correct address in the destination country. The letters will probably travel through one or more international mail gateways along the way, but this is transparent to the users. Furthermore, that each country (i.e., each network) has its own stamps, preferred envelope sizes, and delivery rules is hidden from the users.

The internet layer defines an official packet format and protocol called IP (Internet Protocol), plus a companion protocol called ICMP (Internet Control Message Protocol) that helps it function. The internet layer is to deliver IP packets where they are supposed to go. Packet routing is clearly a major issue here, as is congestion.

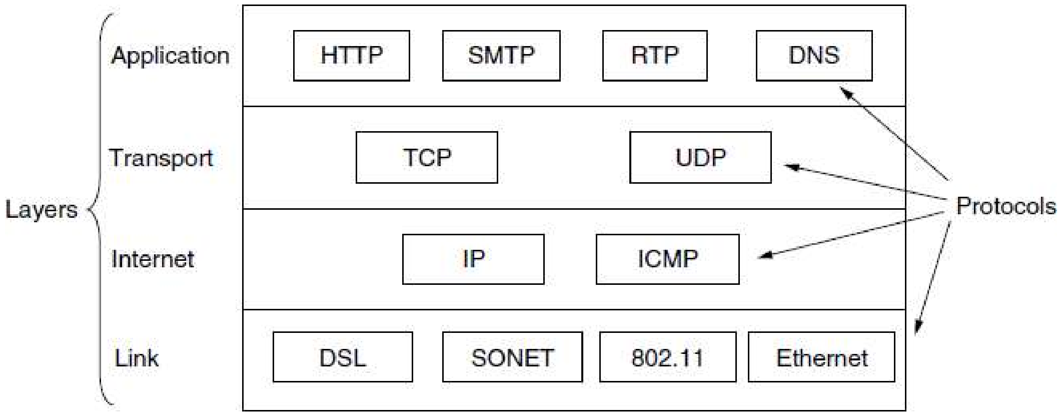
3. The Transport Layer

The layer above the internet layer in the TCP/IP model is now usually called the transport layer. It is designed to allow peer entities on the source and destination hosts to carry on a conversation, just as in the OSI transport layer.

Two end-to-end transport protocols have been defined here.

1. The first one, **Transmission Control Protocol (TCP)**, is a reliable connection-oriented protocol that allows a byte stream originating on one machine to be delivered without error on any other machine in the internet. It segments the incoming byte stream into discrete messages and passes each one on to the internet layer. At the destination, the receiving TCP process reassembles the received messages into the output stream. TCP also handles flow control to make sure a fast sender cannot swamp a slow receiver with more messages than it can handle.
2. The second protocol in this layer, **User Datagram Protocol (UDP)**, is an unreliable, connectionless protocol for applications that do not want TCP's sequencing or flow control and wish to provide their own. It is also widely used for one-shot, client-server-type request-reply queries and

applications in which prompt delivery is more important than accurate delivery, such as transmitting speech or video.



The TCP/IP model with some protocols

4. The Application Layer

The TCP/IP model does not have session or presentation layers. On top of the transport layer is the **application layer**. It contains all the higher- level protocols. The early ones included virtual terminal (TELNET), file transfer (FTP), and electronic mail (SMTP). Many other protocols have been added to these over the years. Some important ones that we will study, shown in the above figure which include the Domain Name System (DNS), for mapping host names onto their network addresses, HTTP, the protocol for fetching pages on the World Wide Web, and RTP, the protocol for delivering real-time media such as voice or movies.

9. Compare and contrast the ISO/OSI reference model with the TCP/IP reference model. (MAY 2017)

The OSI and TCP/IP reference models have much in common. Both are based on the concept of a stack of independent protocols. Also, the functionality of the layers is roughly similar.

For example, in both models the layers up through and including the transport layer are there to provide an end-to-end, network- independent transport service to processes wishing to communicate. These layers form the transport provider. Again in both models, the layers above transport are application-oriented users of the transport service.

Three concepts are central to the OSI model:

1. Services
2. Interfaces
3. Protocols

- Each layer performs some **services** for the layer above it. The service definition tells what the layer does, not how entities above it access it or how the layer works. It defines the layer's semantics.
- A layer's **interface** tells the processes above it how to access it. It specifies what the parameters are and what results to expect.

- Finally, the peer **protocols** used in a layer are the layer's own business. It can use any protocols it wants to, as long as it gets the job done (i.e., provides the offered services). It can also change them at will without affecting software in higher layers.

- The TCP/IP model did not originally clearly distinguish between services, interfaces, and protocols.
- For example, the only real services offered by the internet layer are SEND IP PACKET and RECEIVE IP PACKET.
- The protocols in the OSI model are better hidden than in the TCP/IP model and can be replaced relatively easily as the technology changes.
- The OSI reference model was devised before the corresponding protocols.
- For example, the data link layer originally dealt only with point-to-point networks. When broadcast networks came around, a new sub-layer had to be hacked into the model.

- The difference between the two models is the number of layers: the OSI model has seven layers and the TCP/IP model has four.
- Both have (inter)network, transport, and application layers, but the other layers are different.

Another difference is in the area of connectionless versus connection-oriented communication.

- The OSI model supports both connectionless and connection-oriented communication in the network layer, but only connection-oriented communication in the transport layer, where it counts (because the transport service is visible to the users).
- The TCP/IP model supports only one mode in the network layer (connectionless) but both in the transport layer, giving the users a choice. This choice is especially important for simple request-response protocols.

10. Explain the various examples of networks? (11 MARKS)

The various examples of networks are

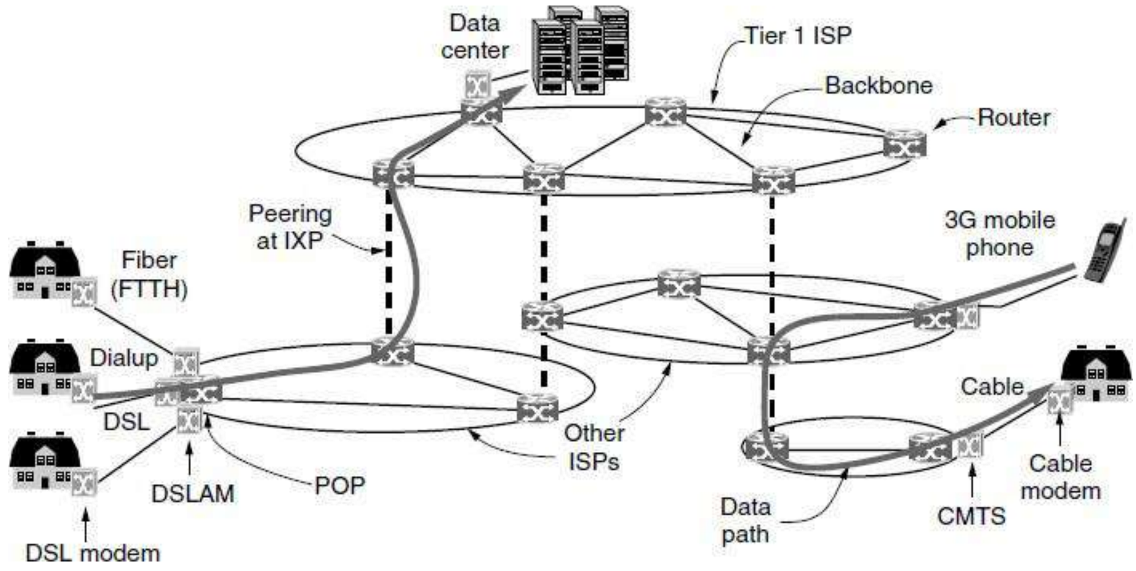
1. The Internet
2. Third-Generation Mobile Phone Networks
3. Wireless LANs: 802.11
4. RFID and Sensor Networks

<u>1. The Internet</u>

The Internet is a network, but a vast collection of different networks that use certain common protocols and provide certain common services.

<u>Architecture of the Internet</u>

Internet, the computer is connected to an Internet Service Provider, or simply ISP, from who the user purchases Internet access or connectivity. The computer exchange packets with all of the other accessible hosts on the Internet. The user might send packets to surf the Web or for any of a thousand other uses, it does not matter. There are many kinds of Internet access, and they are usually distinguished by how much bandwidth they provide and how much they cost, but the most important attribute is connectivity.



Overview of the Internet architecture

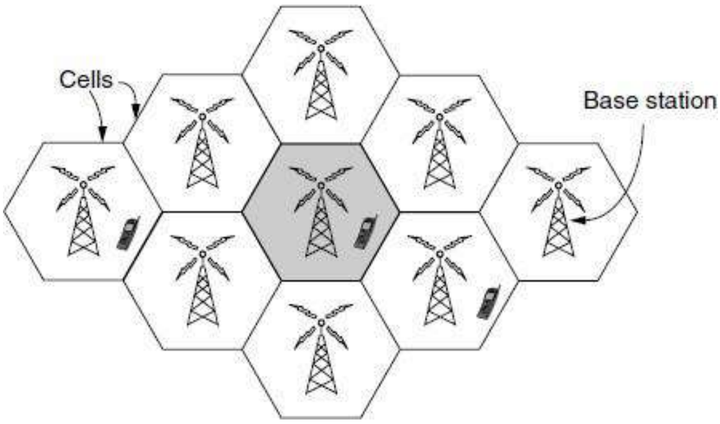
- A common way to connect to an ISP is to use the phone line to your house, in which case your phone company is your ISP.
- **Digital Subscriber Line (DSL)**, reuses the telephone line that connects to your house for digital data transmission. The computer is connected to a device called a DSL modem that converts between digital packets and analog signals that can pass unhindered over the telephone line.
- At the other end, a device called a **Digital Subscriber Line Access Multiplexer (DSLAM)** converts between signals and packets.
- DSL is a higher-bandwidth way to use the local telephone line than to send bits over a traditional telephone call instead of a voice conversation. That is called **dial-up** and done with a different kind of modem at both ends.
- The **modem** (modulator demodulator) refers to any device that converts between digital bits and analog signals.
- Another method is to send signals over the cable TV system. Like DSL, this is a way to reuse existing infrastructure, in this case otherwise unused cable TV channels. The device at the home end is called a **cable modem** and the device at the **cable headend** is called **the CMTS (Cable Modem Termination System).**
- DSL and cable provide Internet access at rates from a small fraction of a megabit/sec to multiple megabit/sec, depending on the system. These rates are much greater than dial-up rates, which are limited to 56 kbps because of the narrow bandwidth used for voice calls. Internet access at much greater than dial-up speeds is called **broadband**. The name refers to the broader bandwidth that is used for faster networks, rather than any particular speed.

- The access methods mentioned so far are limited by the bandwidth of the ''last mile'' or last leg of transmission. By running optical fiber to residences, faster Internet access can be provided at rates on the order of 10 to 100 Mbps. This design is called **FTTH (Fiber to the Home).**
- We can now move packets between the home and the ISP. We call the location at which customer packets enter the ISP network for service the ISP's **POP (Point of Presence).**
- ISP networks may be regional, national, or international in scope. The architecture is made up of long-distance transmission lines that interconnect routers at POPs in the different cities that the ISPs serve. This equipment is called the **backbone** of the ISP. If a packet is destined for a host served directly by the ISP, that packet is routed over the backbone and delivered to the host. Otherwise, it must be handed over to another ISP.
- ISPs connect their networks to exchange traffic at **IXPs (Internet eXchange Points)**. The connected ISPs are said to peer with each other. There are many IXPs in cities around the world.
- They are usually called **tier 1 ISPs** and are said to form the backbone of the Internet, since everyone else must connect to them to be able to reach the entire Internet.
- Companies that provide lots of content, such as Google and Yahoo!, locate their computers in **data centers** that are well connected to the rest of the Internet. These data centers are designed for computers, not humans, and may be filled with rack upon rack of machines called a **server farm.**
- **Colocation or hosting** data centers let customers put equipment such as servers at ISP POPs so that short, fast connections can be made between the servers and the ISP backbones.

2. Third-Generation Mobile Phone Networks

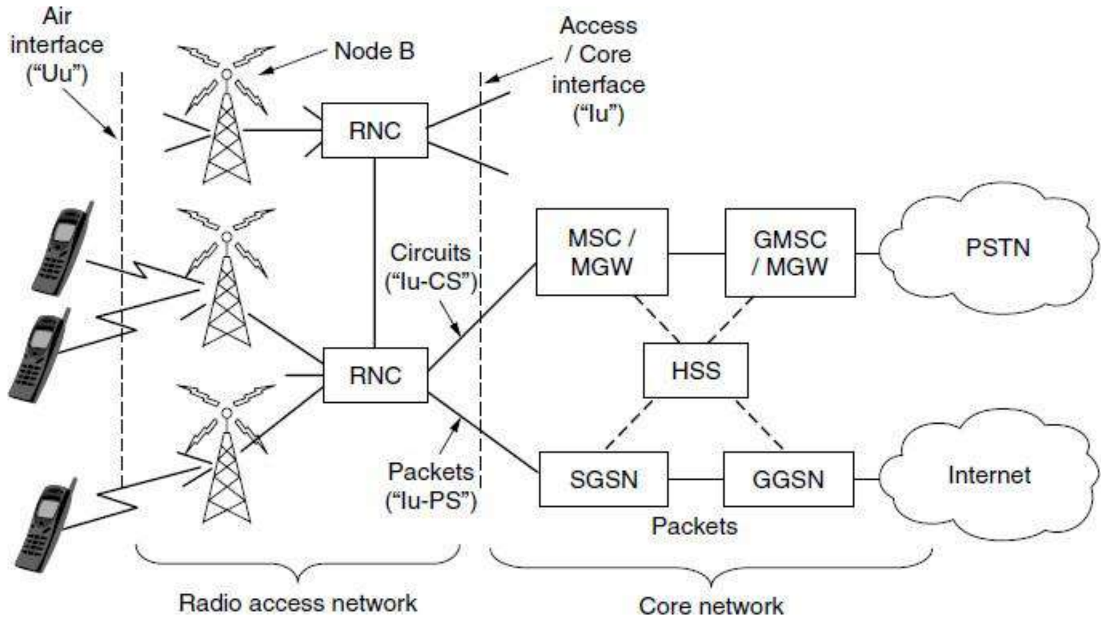
- The architecture of the mobile phone network has changed greatly over the past 40 years along with its tremendous growth.
- First-generation mobile phone systems transmitted voice calls as continuously varying (analog) signals rather than sequences of (digital) bits. **AMPS (Advanced Mobile Phone System)**, which was deployed in the United States in 1982, was a widely used first generation system.
- Second-generation mobile phone systems switched to transmitting voice calls in digital form to increase capacity, improve security, and offer text messaging. **GSM (Global System for Mobile communications)**, which was deployed starting in 1991 and has become the most widely used mobile phone system in the world, is a 2G system.
- The third generation, or 3G, systems were initially deployed in 2001 and offer both digital voice and broadband digital data services. 3G is loosely defined by the ITU as providing rates of at least 2 Mbps for stationary or walking users and 384 kbps in a moving vehicle.
- **UMTS (Universal Mobile Telecommunications System)**, also called **WCDMA (Wideband Code Division Multiple Access)**, is the main 3G system that is being rapidly deployed worldwide. It can provide up to 14 Mbps on the downlink and almost 6 Mbps on the uplink. Future releases will use multiple antennas and radios to provide even greater speeds for users.

- It is the scarcity of spectrum that led to the **cellular network** design shown in Figure that is now used for mobile phone networks. To manage the radio interference between users, the coverage area is divided into cells.
- Within a cell, users are assigned channels that do not interfere with each other and do not cause too much interference for adjacent cells. This allows for good reuse of the spectrum, or **frequency reuse**, in the neighboring cells, which increases the capacity of the network.



Cellular design of mobile phone networks

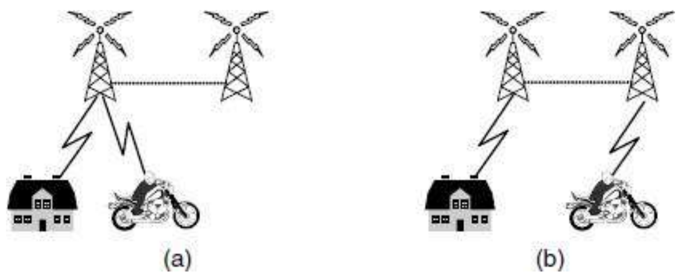
- The architecture of the mobile phone network is very different than that of the Internet. It has several parts, as shown in the simplified version of the UMTS architecture in Figure.
- First, there is the **air interface**. This term is a fancy name for the radio communication protocol that is used over the air between the mobile device (e.g., the cell phone) and the **cellular base station**. Advances in the air interface over the past decades have greatly increased wireless data rates.
- The UMTS air interface is based on **Code Division Multiple Access (CDMA).**
- The cellular base station together with its controller forms the **radio access network**. This part is the wireless side of the mobile phone network. The controller node or **RNC (Radio Network Controller)** controls how the spectrum is used. The base station implements the air interface. It is called **Node B**, a temporary label that stuck.
- The rest of the mobile phone network carries the traffic for the radio access network. It is called the **core network.** The UMTS core network evolved from the core network used for the 2G GSM system that came before it.



Architecture of the UMTS 3G mobile phone network

- The UMTS network with the **MSC (Mobile Switching Center), GMSC (Gate-way Mobile Switching Center),** and **MGW (Media Gateway)** elements that set up connections over a circuit-switched core network such as the **PSTN (Public Switched Telephone Network).**
- Data services have become a much more important part of the mobile phone network than they used to be, starting with text messaging and early packet data services such as **GPRS (General Packet Radio Service)** in the GSM system.
- These older data services ran at tens of kbps, but users wanted more. Newer mobile phone networks carry packet data at rates of multiple Mbps. For comparison, a voice call is carried at a rate of 64 kbps, typically 3–4x less with compression.
- To carry all this data, the UMTS core network nodes connect directly to a packet-switched network. The **SGSN (Serving GPRS Support Node)** and the **GGSN (Gateway GPRS Support Node)** deliver data packets to and from mobiles and interface to external packet networks such as the Internet.

Another difference between mobile phone networks and the traditional Internet is mobility. When a user moves out of the range of one cellular base station and into the range of another one, the flow of data must be re-routed from the old to the new cell base station. This technique is known as **handover or handoff.**



Mobile phone handover (a) before, (b) after

Either the mobile device or the base station may request a handover when the quality of the signal drops. In some cell networks, usually those based on CDMA technology, it is possible to connect to the new base station before disconnecting from the old base station.

This improves the connection quality for the mobile because there is no break in service; the mobile is actually connected to two base stations for a short while. This way of doing a handover is called a **soft handover t**o distinguish it from a **hard handover**, in which the mobile disconnects from the old base station before connecting to the new one.

A related issue is how to find a mobile in the first place when there is an incoming call. Each mobile phone network has a **HSS (Home Subscriber Server)** in the core network that knows the location of each subscriber, as well as other profile information that is used for authentication and authorization.

Starting with the 2G GSM system, the mobile phone was divided into a handset and a removable chip containing the subscriber's identity and account information. The chip is informally called a **SIM card, (Subscriber Identity Module)**. SIM cards can be switched to different handsets to activate them, and they provide a basis for security. When GSM customers travel to other countries on vacation or business, they often bring their handsets but buy a new SIM card for few dollars upon arrival in order to make local calls with no roaming charges.

To reduce fraud, information on SIM cards is also used by the mobile phone network to authenticate subscribers and check that they are allowed to use the network. With UMTS, the mobile also uses the information on the SIM card to check that it is talking to a legitimate network.

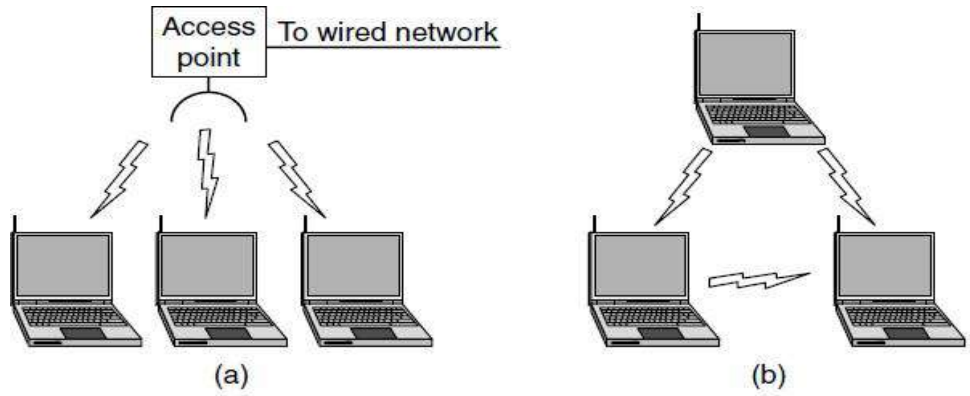
Another aspect of **security** is privacy. Wireless signals are broadcast to all nearby receivers, so to make it difficult to eavesdrop on conversations; cryptographic keys on the SIM card are used to encrypt transmissions. This approach provides much better privacy than in 1G systems, which were easily tapped.

Mobile phone networks are destined to play a central role in future networks. They are now more about mobile broadband applications than voice calls, and this has major implications for the air interfaces, core network architecture, and security of future networks. 4G technologies that are faster and better are on the drawing board under the name of **LTE (Long Term Evolution),** even as 3G design and deployment continues.

Other wireless technologies also offer broadband Internet access to fixed and mobile clients, notably 802.16 networks under the common name of **WiMAX.**

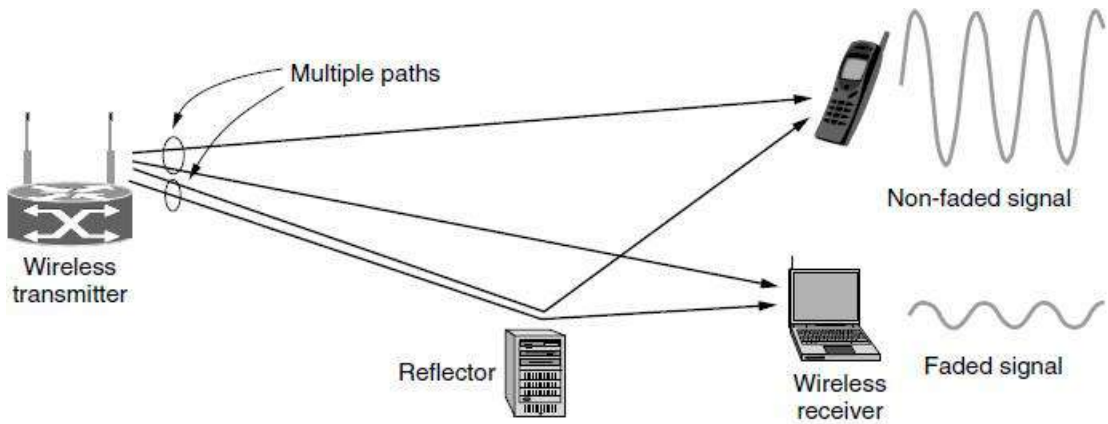
3. Wireless LANs: 802.11

- 802.11 networks are made up of clients, such as laptops and mobile phones, and infrastructure called **APs (access points)** that is installed in buildings. Access points are sometimes called **base stations.**
- The access points connect to the wired network, and all communication between clients goes through an access point.
- It is also possible for clients that are in radio range to talk directly, such as two computers in an office without an access point. This arrangement is called an **ad hoc network.**
- It is used much less often than the access point mode.



(a) Wireless network with an access point. (b) Ad hoc network.

802.11 transmission is complicated by wireless conditions that vary with even small changes in the environment. At the frequencies used for 802.11, radio signals can be reflected off solid objects so that multiple echoes of a transmission may reach a receiver along different paths. The echoes can cancel or reinforce each other, causing the received signal to fluctuate greatly. This phenomenon is called **multipath fading.**



Multipath fading.

The key idea for overcoming variable wireless conditions is **path diversity**, or the sending of information along multiple, independent paths. In this way, the information is likely to be received even if one of the paths happens to be poor due to a fade. These independent paths are typically built into the digital modulation scheme at the physical layer. Options include using different frequencies across the allowed band, following different spatial paths between different pairs of antennas, or repeating bits over different periods of time.

Different versions of 802.11 have used all of these techniques.

- The initial (1997) standard defined a wireless LAN that ran at either 1 Mbps or 2 Mbps by hopping between frequencies or spreading the signal across the allowed spectrum. Almost immediately, people complained that it was too slow, so work began on faster standards.
- The spread spectrum design was extended and became the (1999) 802.11b standard running at rates up to 11 Mbps.

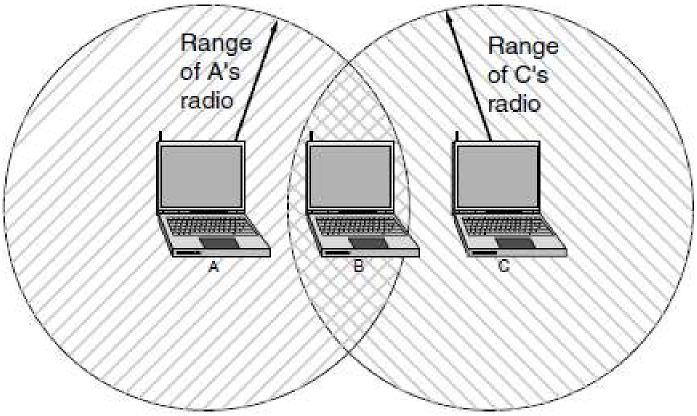
- The 802.11a (1999) and 802.11g (2003) standards switched to a different modulation scheme called **OFDM (Orthogonal Frequency Division Multiplexing).** It divides a wide band of spectrum into many narrow slices over which different bits are sent in parallel.
- The 802.11a/g bit rates up to 54 Mbps. That is a significant increase, but people still wanted more throughput to support more demanding uses.
- The latest version is 802.11n (2009). It uses wider frequency bands and up to four antennas per computer to achieve rates up to 450 Mbps.

Since wireless is inherently a broadcast medium, 802.11 radios also have to deal with the problem that multiple transmissions that are sent at the same time will collide, which may interfere with reception. To handle this problem, 802.11 uses a **CSMA (Carrier Sense Multiple Access)** scheme that draws on ideas from classic wired Ethernet, which, ironically, drew from an early wireless network developed in Hawaii and called **ALOHA.**

Computers wait for a short random interval before transmitting, and defer their transmissions if they hear that someone else is already transmitting. This scheme makes it less likely that two computers will send at the same time. It does not work as well as in the case of wired networks, though.

Suppose that computer A is transmitting to computer B, but the radio range of A's transmitter is too short to reach computer C. If C wants to transmit to B it can listen before starting, but the fact that it does not hear anything does not mean that its transmission will succeed. The inability of C to hear A before starting causes some collisions to occur. After any collision, the sender then waits another, longer, random delay and retransmits the packet.

- Another problem is that of mobility. If a mobile client is moved away from the access point it is using and into the range of a different access point, some way of handing it off is needed.
- The solution is that an 802.11 network can consist of multiple cells, each with its own access point, and a distribution system that connects the cells.
- The distribution system is often switched Ethernet, but it can use any technology. As the clients move, they may find another access point with a better signal than the one they are currently using and change their association. From the outside, the entire system looks like a single wired LAN.

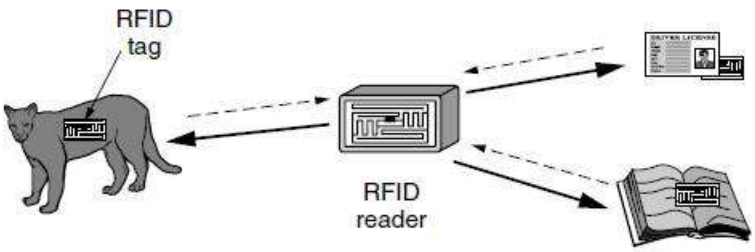


The range of a single radio may not cover the entire system.

- Mobility in 802.11 has been of limited value so far compared to mobility in the mobile phone network. Typically, 802.11 is used by nomadic clients that go from one fixed location to another, rather than being used on-the-go. Mobility is not really needed for nomadic usage.
- Even when 802.11 mobility is used, it extends over a single 802.11 network, which might cover at most a large building.
- Finally, there is the problem of security. Since wireless transmissions are broadcast, it is easy for nearby computers to receive packets of information that were not intended for them. To prevent this, the 802.11 standard included an encryption scheme known as **WEP (Wired Equivalent Privacy).**
- The idea was to make wireless security like that of wired security. It has since been replaced with newer schemes that have different cryptographic details in the 802.11i standard, also called **WiFi Protected Access**, initially called **WPA** but now replaced by WPA2.

4. RFID and Sensor Networks

- **Radio Frequency IDentification (RFID)**, everyday objects can also be part of a computer network.
- An RFID tag looks like a postage stamp-sized sticker that can be affixed to (or embedded in) an object so that it can be tracked. The object might be a cow, a passport, a book or a shipping pallet. The tag consists of a small microchip with a unique identifier and an antenna that receives radio transmissions. RFID readers installed at tracking points find tags when they come into range and interrogate them for their information.



RFID used to network everyday objects.

Applications include

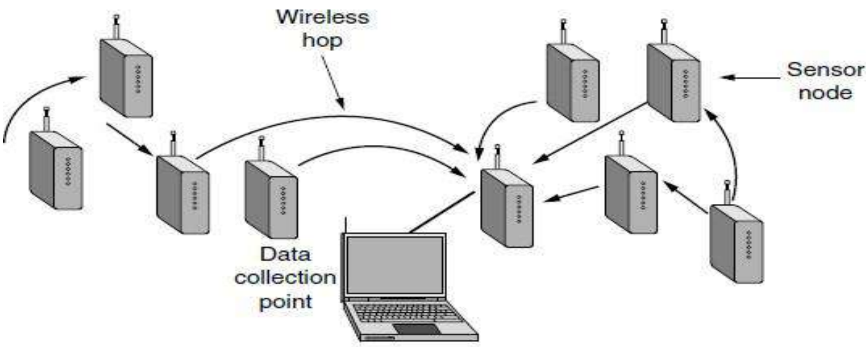
- checking identities,
- managing the supply chain,

- timing races, and
- replacing barcodes

There are many kinds of RFID, each with different properties

1. The energy needed to operate them is supplied in the form of radio waves by RFID readers. This technology is called **passive RFID** to distinguish it from the (less common) **active RFID** in which there is a power source on the tag.
2. One common form of RFID is **UHF RFID (Ultra-High Frequency RFID).** It is used on shipping pallets and some drivers licenses. Readers send signals in the 902-928 MHz band in the United States. Tags communicate at distances of several meters by changing the way they reflect the reader signals; the reader is able to pick up these reflections. This way of operating is called **backscatter.**
3. Another popular kind of RFID is **HF RFID (High Frequency RFID)**. It operates at 13.56 MHz and is likely to be in your passport, credit cards, books, and noncontact payment systems.
4. There are also other forms of RFID using other frequencies, such as **LF RFID (Low Frequency RFID)**, which was developed before HF RFID and used for animal tracking.

- A step up in capability from RFID is the **sensor network**. Sensor networks are deployed to monitor aspects of the physical world.
- The mostly been used for scientific experimentation, such as
  - monitoring bird habitats,
  - volcanic activity, and
  - zebra migration,
  - business applications including healthcare,
  - monitoring equipment for vibration, and
  - tracking of frozen,
  - refrigerated
- Sensor nodes are small computers, often the size of a key fob, that have temperature, vibration, and other sensors.
- Many nodes are placed in the environment that is to be monitored. Typically, they have batteries, though they may scavenge energy from vibrations or the sun.



Multihop topology of a sensor network.

- As with RFID, having enough energy is a key challenge, and the nodes must communicate carefully to be able to deliver their sensor information to an external collection point.

- A common strategy is for the nodes to self-organize to relay messages for each other, as shown in Figure. This design is called a **multihop network.**

11. Explain the theoretical basis for Data Communication? (11 MARKS) (NOV 2015)

Information can be transmitted on wires by varying some physical property such as voltage or current. By representing the value of this voltage or current as a single-valued function of time, f(t), we can model the behavior of the signal and analyze it mathematically.

Fourier analysis:

In the early 19th century, the French mathematician Jean-Baptiste Fourier proved that any reasonably behaved periodic function, g(t) with period T, can be constructed as the sum of a (possibly infinite) number of sines and cosines:

$$g(t) = \frac{1}{2}c + \sum_{n=1}^{\infty} a_n \sin(2\pi nft) + \sum_{n=1}^{\infty} b_n \cos(2\pi nft) \qquad (2\text{-}1)$$

Where f =1/T is the fundamental frequency, a_n and b_n are the sine and cosine amplitudes of the nth harmonics (terms), and c is a constant. Such a decomposition is called a Fourier series. From the Fourier series, the function can be reconstructed. That is, if the period, T, is known and the amplitudes are given, the original function of time can be found by performing the sums of Eq. (2-1).

A data signal that has a finite duration, which all of them do, can be handled by just imagining that it repeats the entire pattern over and over forever (i.e., the interval from T to 2T is the same as from 0 to T, etc.).

The an amplitudes can be computed for any given g(t) by multiplying both sides of Eq. (2-1) by sin(2πkft) and then integrating from 0 to T.

$$\int_0^T \sin(2\pi kft)\sin(2\pi nft)\,dt = \begin{cases} 0 \text{ for } k \neq n \\ T/2 \text{ for } k = n \end{cases}$$

Since only one term of the summation survives: a_n. The b_n summation vanishes completely. Similarly, by multiplying Eq. (2-1) by cos(2πkft) and integrating between 0 and T, we can derive b_n. By just integrating both sides of the equation as it stands, we can find c. The results of performing these operations are as follows:

$$a_n = \frac{2}{T}\int_0^T g(t)\sin(2\pi nft)\,dt \qquad b_n = \frac{2}{T}\int_0^T g(t)\cos(2\pi nft)\,dt \qquad c = \frac{2}{T}\int_0^T g(t)\,dt$$

Bandwidth-Limited Signals

Let us consider a specific example: the transmission of the ASCII character "b" encoded in an 8-bit byte. The bit pattern that is to be transmitted is 01100010. The left-hand part of Figure shows the voltage output by the transmitting computer. The Fourier analysis of this signal yields the coefficients:

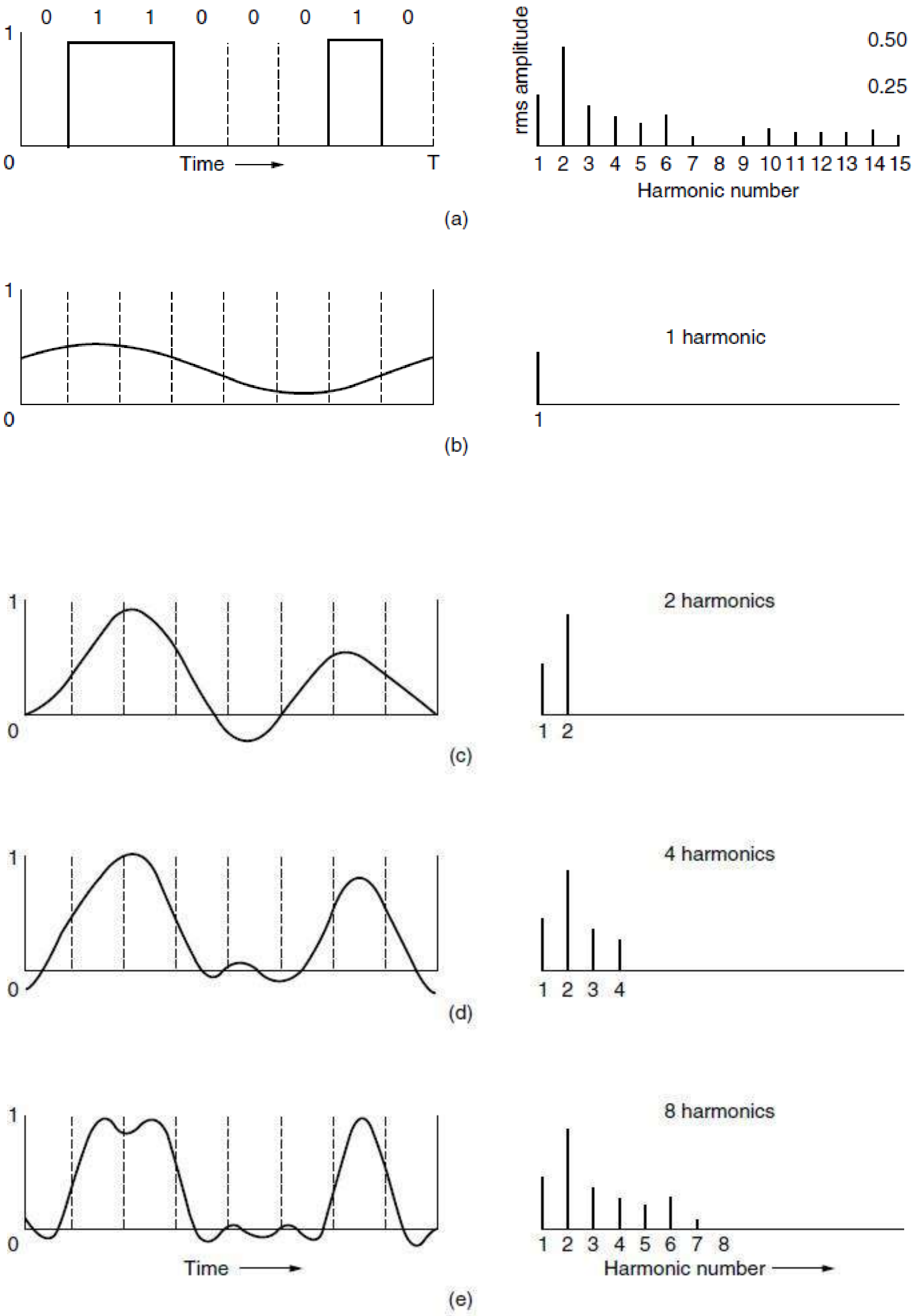
$$a_n = \frac{1}{\pi n}[\cos(\pi n/4) - \cos(3\pi n/4) + \cos(6\pi n/4) - \cos(7\pi n/4)]$$

$$b_n = \frac{1}{\pi n}[\sin(3\pi n/4) - \sin(\pi n/4) + \sin(7\pi n/4) - \sin(6\pi n/4)]$$

$$c = 3/4$$

The root-mean-square amplitudes, $\sqrt{a^2_n + b^2_n}$, for the first few terms are shown on the right-hand side of Figure. These values are of interest because their squares are proportional to the energy transmitted at the corresponding frequency.

- No transmission facility can transmit signals without losing some power in the process. If all the Fourier components were equally diminished, the resulting signal would be reduced in amplitude but not distorted [i.e., it would have the same nice squared-off shape as Fig. 2-1(a)].
- Unfortunately, all transmission facilities diminish different Fourier components by different amounts, thus introducing distortion. Usually, for a wire, the amplitudes are transmitted mostly undiminished from 0 up to some frequency fc [measured in cycles/sec or Hertz (Hz)], with all frequencies above this cutoff frequency attenuated.
- The width of the frequency range transmitted without being strongly attenuated is called the **bandwidth.** In practice, the cutoff is not really sharp, so often the quoted bandwidth is from 0 to the frequency at which the received power has fallen by half.
- The bandwidth is a physical property of the transmission medium that depends on, for example, the construction, thickness, and length of a wire or fiber. Filters are often used to further limit the bandwidth of a signal. 802.11 wireless channels are allowed to use up to roughly 20 MHz, for example, so 802.11 radios filter the signal bandwidth to this size.
- As another example, traditional (analog) television channels occupy 6 MHz each, on a wire or over the air. This filtering lets more signals share a given region of spectrum, which improves the overall efficiency of the system. It means that the frequency range for some signals will not start at zero, but this does not matter. The bandwidth is still the width of the band of frequencies that are passed, and the information that can be carried depends only on this width and not on the starting and ending frequencies.
- Signals that run from 0 up to a maximum frequency are called **baseband signals.**
- Signals that are shifted to occupy a higher range of frequencies, as is the case for all wireless transmissions, are called **passband signals.**



(a) A binary signal and its root-mean-square Fourier amplitudes. (b)–(e) Successive approximations to the original signal.

The Maximum Data Rate of a Channel

- Nyquist proved that if an arbitrary signal has been run through a low-pass filter of bandwidth B, the filtered signal can be completely reconstructed by making only 2B (exact) samples per second.
- Sampling the line faster than 2B times per second is pointless because the higher-frequency components that such sampling could recover have already been filtered out.

If the signal consists of V discrete levels, **Nyquist's theorem** states:

$$\textbf{maximum data rate} = 2B \log_2 V \textbf{ bits /sec}$$

For example, a noiseless 3-kHz channel cannot transmit binary (i.e., two-level) signals at a rate exceeding 6000 bps. The amount of thermal noise present is measured by the ratio of the signal power to the noise power, called the **SNR (Signal-to-Noise Ratio). Shannon's** major result is that the maximum data rate or capacity of a noisy channel whose bandwidth is B Hz and whose signal-to-noise ratio is S/N, is given by:

$$\textbf{maximum number of bits/sec} = B \log_2 (1+S/N)$$

12. Explain in detail about transmission guided media? (11 MARKS) (NOV 2015) (MAY 2016, 2017)

- The purpose of the physical layer is to transport bits from one machine to another.
- Various physical media can be used for the actual transmission.
- Each one has its own niche in terms of bandwidth, delay, cost, and ease of installation and maintenance.

Media are roughly grouped into

1. **Guided media,** such as
   - copper wire and
   - fiber optics
2. **Unguided media,** such as
   - terrestrial wireless,
   - satellite, and
   - lasers through the air

Guided Media:

Guided media, which are those that provide a conduit from one device to another, include

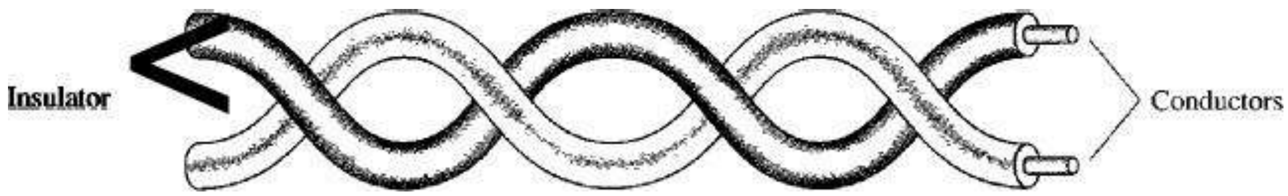
- Magnetic media,
- Twisted-pair cable,
- Coaxial cable,
- Power Lines,
- Fiber-optic cable

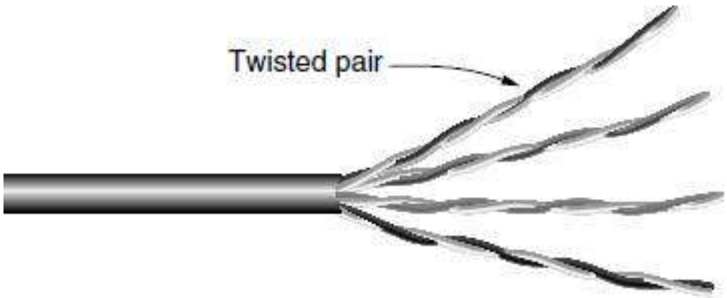
1. Magnetic Media

- One of the most common ways to transport data from one computer to another is to write them onto **magnetic tape or removable media** (e.g., recordable DVDs), physically transport the tape or disks to the destination machine, and read them back in again.
- Although this method is not as sophisticated as using a geosynchronous communication satellite, it is often more cost effective, especially for applications in which high bandwidth or cost per bit transported is the key factor.

2. Twisted Pairs

Although the bandwidth characteristics of magnetic tape are excellent, the delay characteristics are poor. Transmission time is measured in minutes or hours. One of the oldest and still most common transmission media is **twisted pair**. A twisted pair consists of two insulated copper wires, typically about 1 mm thick. The wires are twisted together in a helical form, just like a DNA molecule. Twisting is done because two parallel wires constitute a fine antenna. When the wires are twisted, the waves from different twists cancel out, so the wire radiates less effectively.





Category 5 UTP cable with four twisted pairs.

Twisted-pair cabling comes in several varieties. The garden variety deployed in many office buildings is called **Category 5** cabling, or ''Cat 5.'' A category 5 twisted pair consists of two insulated wires gently twisted together. Four such pairs are typically grouped in a plastic sheath to protect the wires and keep them together. This arrangement is shown in above figure.

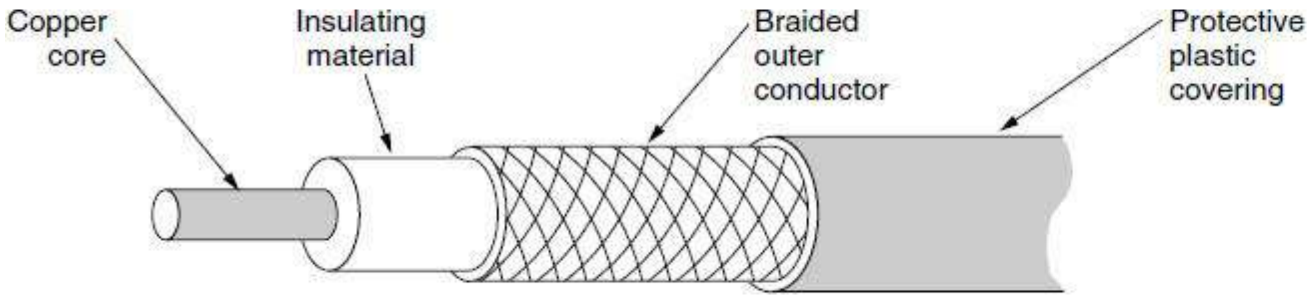
In twisted pair, Cat 5 replaced earlier **Category 3** cables with a similar cable that uses the same connector, but has more twists per meter. More twists result in less crosstalk and a better-quality signal over longer distances, making the cables more suitable for high-speed computer communication, especially 100-Mbps and 1-Gbps Ethernet LANs.

New wiring is more likely to be **Category 6** or even **Category 7**. These categories have more stringent specifications to handle signals with greater bandwidths. Some cables in Category 6 and above are rated for signals of 500 MHz and can support the 10-Gbps links that will soon be deployed. Through Category 6, these wiring types are referred to as **UTP** (**Unshielded Twisted Pair**) as they consist simply of wires and insulators. In contrast to these, Category 7 cables have shielding on the individual twisted pairs, as well as around the entire cable (but inside the plastic protective sheath). Shielding reduces the susceptibility to external interference and crosstalk with other nearby cables to meet demanding performance specifications. The cables are reminiscent of the high-quality, but bulky and expensive shielded

3. Coaxial Cable

- Another common transmission medium is the coaxial cable. It has better shielding and greater bandwidth than unshielded twisted pairs, so it can span longer distances at higher speeds.
- Two kinds of coaxial cable are widely used.
  1. One kind, 50-ohm cable, is commonly used when it is intended for digital transmission from the start.
  2. The other kind, 75-ohm cable, is commonly used for analog transmission and cable television.

A coaxial cable consists of a stiff copper wire as the core, surrounded by an insulating material. The insulator is encased by a cylindrical conductor, often as a closely woven braided mesh. The outer conductor is covered in a protective plastic sheath.



A coaxial cable

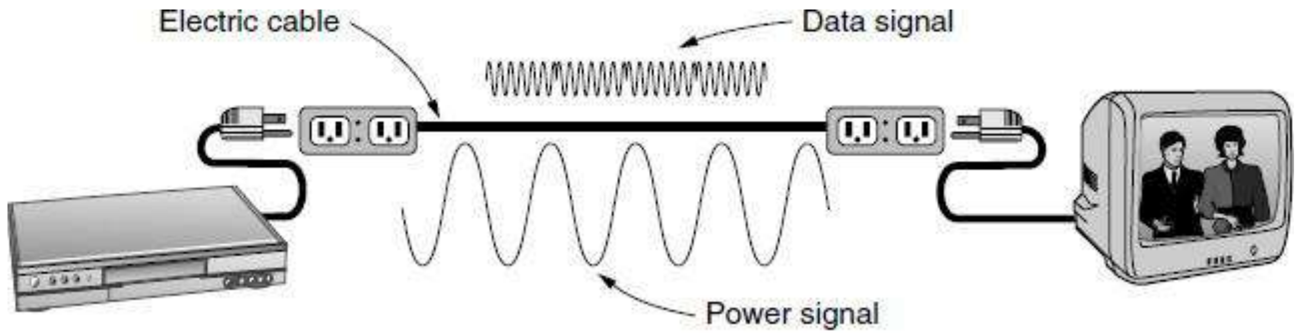
The construction and shielding of the coaxial cable give it a good combination of high bandwidth and excellent noise immunity.

- The bandwidth possible depends on the cable quality and length. Modern cables have a bandwidth of up to a few GHz.
- Coaxial cables used to be widely used within the telephone system for long-distance lines but have now largely been replaced by fiber optics on long-haul routes.
- Coax is still widely used for cable television and metropolitan area networks.

4. Power Lines

Power lines have been used by electricity companies for low-rate communication such as remote metering for many years, as well in the home to control devices.

The convenience of using power lines for networking should be clear. Simply plug a TV and a receiver into the wall, which you must do anyway because they need power, and they can send and receive movies over the electrical wiring. This configuration is shown in Figure.



A network that uses household electrical wiring.

The difficulty with using household electrical wiring for a network is that it was designed to distribute power signals. This task is quite different than distributing data signals, at which household wiring does a horrible job. Electrical signals are sent at 50–60 Hz and the wiring attenuates the much higher frequency (MHz) signals needed for high-rate data communication.

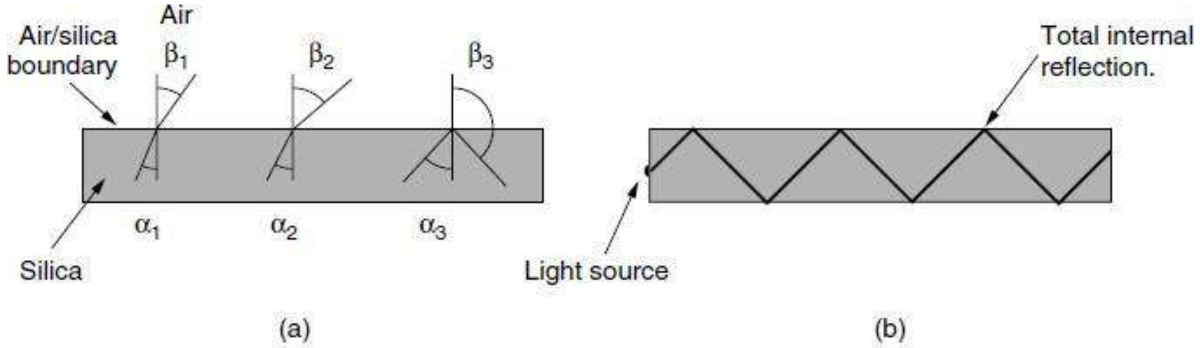
5. Fiber Optics

A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.
Fiber optics are used for long-haul transmission in network backbones, high speed LANs and high-speed Internet access such as **FttH (Fiber to the Home).**
An optical transmission system has three key components:

- the light source,
- the transmission medium, and
- the detector

Conventionally, a pulse of light indicates a 1 bit and the absence of light indicates a 0 bit. The transmission medium is an ultra-thin fiber of glass. The detector generates an electrical pulse when light falls on it. By attaching a light source to one end of an optical fiber and a detector to the other, we have a unidirectional data transmission system that accepts an electrical signal, converts and transmits it by light pulses, and then reconverts the output to an electrical signal at the receiving end.
This transmission system would leak light and be useless in practice were it not for an interesting principle of physics.
The Figure. 2-6(b) shows only one trapped ray, but since any light ray incident on the boundary above the critical angle will be reflected internally, many different rays will be bouncing around at different angles. Each ray is said to have a different mode, so a fiber having this property is called a **multimode fiber.**

However, if the fiber's diameter is reduced to a few wavelengths of light the fiber acts like a wave guide and the light can propagate only in a straight line, without bouncing, yielding a **single-mode fiber**.

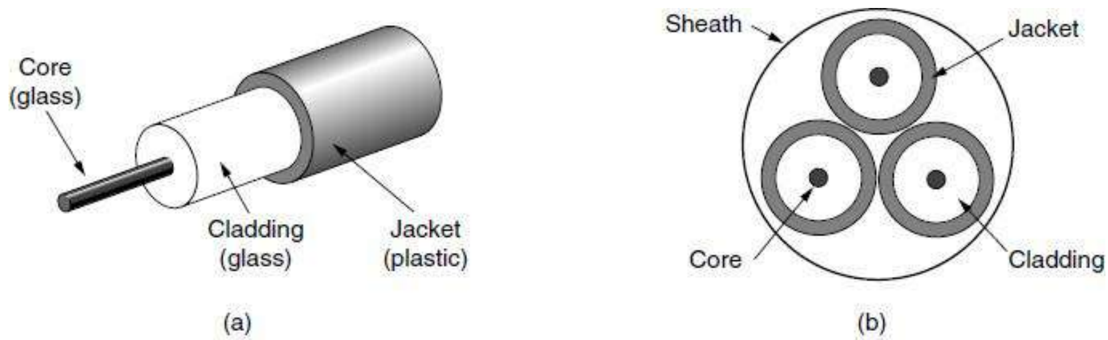


(a) Three examples of a light ray from inside a silica fiber impinging on the air/silica boundary at different angles. (b) Light trapped by total internal reflection.

Single-mode fibers are more expensive but are widely used for longer distances. Currently available single-mode fibers can transmit data at 100 Gbps for 100 km without amplification. Even higher data rates have been achieved in the laboratory for shorter distances.

Fiber Cables

Fiber optic cables are similar to coax, except without the braid. Figure (a) shows a single fiber viewed from the side. At the center is the glass core through which the light propagates. In multimode fibers, the core is typically 50 microns in diameter, about the thickness of a human hair. In single-mode fibers, the core is 8 to 10 microns.



(a) Side view of a single fiber. (b) End view of a sheath with three fibers.

The core is surrounded by a glass cladding with a lower index of refraction than the core, to keep all the light in the core. Next comes a thin plastic jacket to protect the cladding. Fibers are typically grouped in bundles, protected by an outer sheath. Figure (b) shows a sheath with three fibers.

- **Terrestrial fiber sheaths** are normally laid in the ground within a meter of the surface, where they are occasionally subject to attacks by backhoes or gophers.
- Near the shore, **transoceanic fiber sheaths** are buried in trenches by a kind of sea plow. In deep water, they just lie on the bottom, where they can be snagged by fishing trawlers or attacked by giant squid.

Two kinds of light sources are typically used to do the signaling.

These are

- LEDs (Light Emitting Diodes) and

- Semiconductor lasers

A comparison of semiconductor diodes and LEDs as light sources

Item	LED	Semiconductor laser
Data rate	Low	High
Fiber type	Multi-mode	Multi-mode or single-mode
Distance	Short	Long
Lifetime	Long life	Short life
Temperature sensitivity	Minor	Substantial
Cost	Low cost	Expensive

13. Explain in detail about wireless transmission? (11 MARKS) (MAY 2017)

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is often referred to as **wireless communication**. Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.

The various wireless communications are

1. The Electromagnetic Spectrum
2. Radio Transmission
3. Microwave Transmission
4. Infrared Transmission
5. Light Transmission

1. The Electromagnetic Spectrum

- When electrons move, they create electromagnetic waves that can propagate through space (even in a vacuum).
- The number of oscillations per second of a wave is called its **frequency,** f, and is measured in Hz.
- The distance between two consecutive maxima (or minima) is called the **wavelength λ** (lambda).

When an antenna of the appropriate size is attached to an electrical circuit, the electromagnetic waves can be broadcast efficiently and received by a receiver some distance away. All wireless communication is based on this principle.

In a vacuum, all electromagnetic waves travel at the same speed, no matter what their frequency. This speed, usually called the **speed of light**, c, is approximately 3 ×108 m/sec, or about 1 foot (30 cm) per nanosecond. In copper or fiber the speed slows to about 2/3 of this value and becomes slightly frequency dependent. The speed of light is the ultimate speed limit. No object or signal can ever move faster than it.

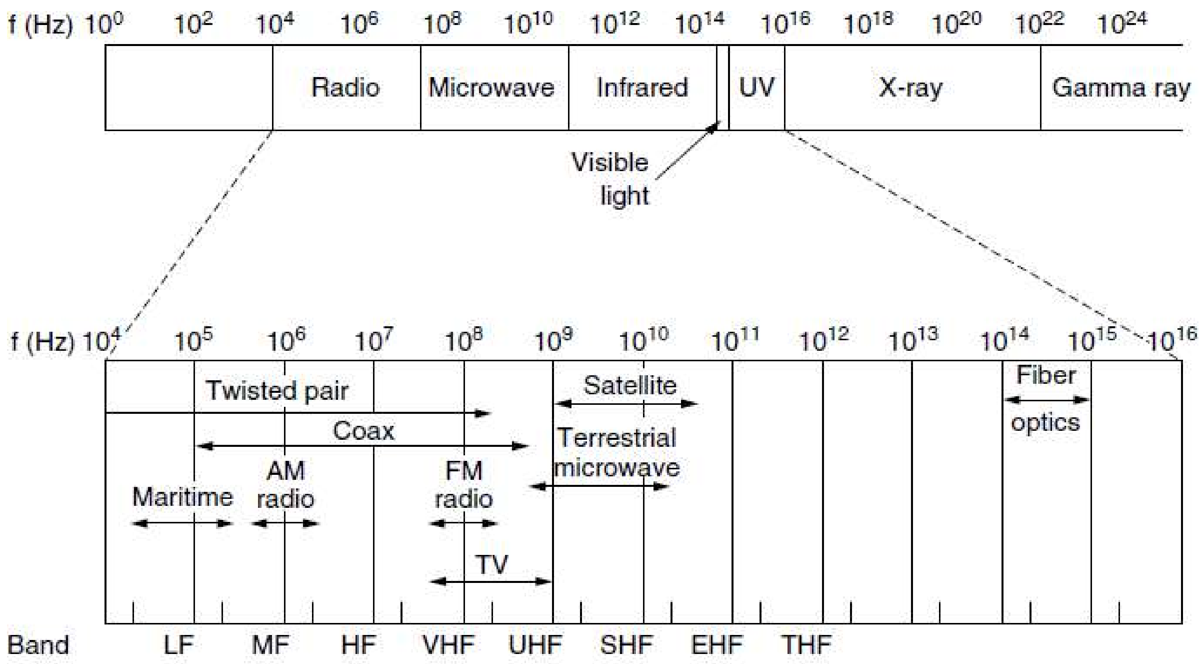
The fundamental relation between f, λ, and c (in a vacuum) is

$$\lambda f = c$$

Since c is a constant, if we know f, we can find λ, and vice versa. As a rule of thumb, when λ is in meters and f is in MHz, $\lambda f = 300$. For example, 100-MHz waves are about 3 meters long, 1000-MHz waves are 0.3 meters long, and 0.1-meter waves have a frequency of 3000 MHz.

The electromagnetic spectrum is shown in Figure.

- The radio, microwave, infrared, and visible light portions of the spectrum can all be used for transmitting information by modulating the amplitude, frequency, or phase of the waves.
- Ultraviolet light, X-rays, and gamma rays would be even better, due to their higher frequencies, but they are hard to produce and modulate, do not propagate well through buildings, and are dangerous to living things.
- The bands listed at the bottom of Figure are the official **ITU (International Telecommunication Union)** names and are based on the wavelengths, so the LF band goes from 1 km to 10 km (approximately 30 kHz to 300 kHz).
- The terms LF, MF, and HF refer to Low, Medium, and High Frequency, respectively.



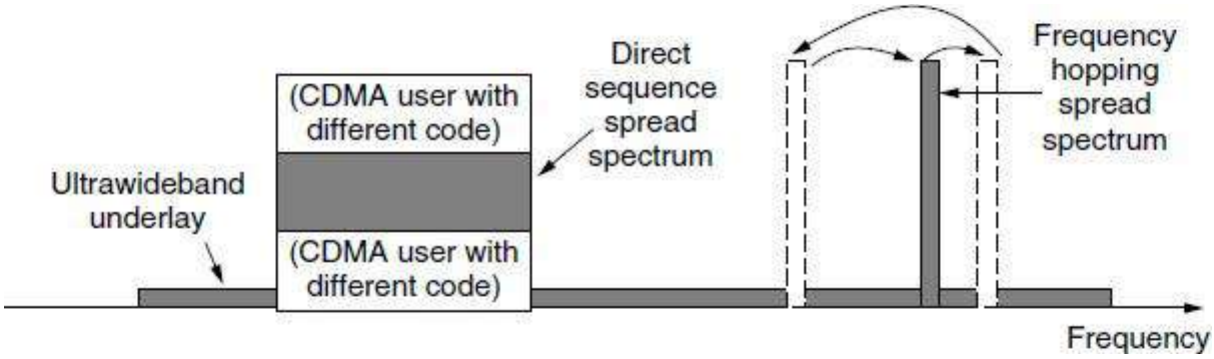
The electromagnetic spectrum and its uses for communication.

Band:

- In **frequency hopping spread spectrum**, the transmitter hops from frequency to frequency hundreds of times per second. This technique is used commercially, for example, in Bluetooth and older versions of 802.11.
- A second form of spread spectrum, **direct sequence spread spectrum**, uses a code sequence to spread the data signal over a wider frequency band. It is widely used commercially as a spectrally efficient way to let multiple signals share the same frequency band. These signals can be given different codes, a method called **CDMA (Code Division Multiple Access).**

Band	*Range*	*Propagation*	*Application*
VLF (very low frequency)	3-30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30-300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz-3 MHz	Sky	AM radio
HF (high frequency)	3-30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30-300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz-3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3-30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30-300 GHz	Line-of-sight	Radar, satellite

This method is shown in Figure contrast with frequency hopping are



Spread spectrum and ultra-wideband (UWB) communication

- It forms the basis of 3G mobile phone networks and is also used in GPS (Global Positioning System).
- Even without different codes, direct sequence spread spectrum, like frequency hopping spread spectrum, can tolerate narrowband interference and multipath fading because only a fraction of the desired signal is lost. It is used in this role in older 802.11b wireless LANs.
- A third method of communication with a wider band is **UWB (Ultra - Wide Band) communication**. UWB sends a series of rapid pulses, varying their positions to communicate information. The rapid transitions lead to a signal that is spread thinly over a very wide frequency band.
- UWB is defined as signals that have a bandwidth of at least 500 MHz or at least 20% of the center frequency of their frequency band.

2. Radio Transmission

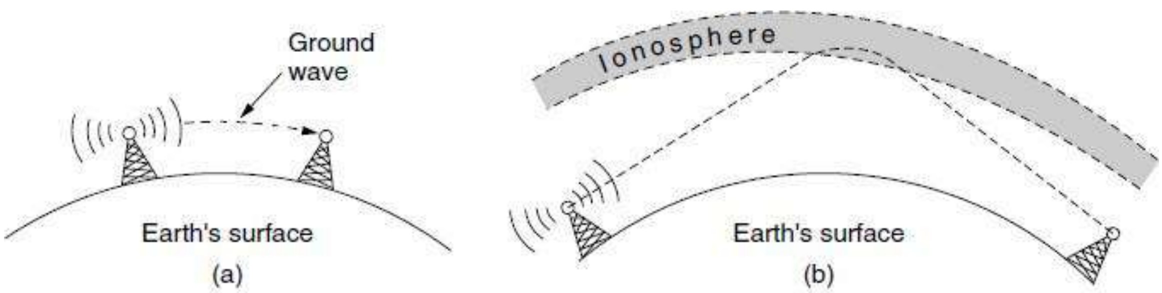
- Radio frequency (RF) waves are easy to generate, can travel long distances, and can penetrate buildings easily, so they are widely used for communication, both indoors and outdoors.

- Radio waves also are omnidirectional, meaning that they travel in all directions from the source, so the transmitter and receiver do not have to be carefully aligned physically.

The properties of radio waves are frequency dependent.

- At low frequencies, radio waves pass through obstacles well, but the power falls off sharply with distance from the source—at least as fast as $1/r^2$ in air—as the signal energy is spread more thinly over a larger surface. This attenuation is called **path loss**.
- At high frequencies, radio waves tend to travel in straight lines and bounce off obstacles. Path loss still reduces power, though the received signal can depend strongly on reflections as well. High-frequency radio waves are also absorbed by rain and other obstacles to a larger extent than are low-frequency ones.
- At all frequencies, radio waves are subject to interference from motors and other electrical equipment.

In the VLF, LF, and MF bands, radio waves follow the ground, as illustrated in Figure (a). These waves can be detected for perhaps 1000 km at the lower frequencies, less at the higher ones. Radio waves in these bands pass through buildings easily, which is why portable radios work indoors. The main problem with using these bands for data communication is their low bandwidth.



(a) In the VLF, LF, and MF bands, radio waves follow the curvature of the earth. (b) In the HF band, they bounce off the ionosphere.

In the HF and VHF bands, the ground waves tend to be absorbed by the earth. However, the waves that reach the ionosphere, a layer of charged particles circling the earth at a height of 100 to 500 km, are refracted by it and sent back to earth, as shown in Fig. 2-12(b).

Under certain atmospheric conditions, the signals can bounce several times. Amateur radio operators (hams) use these bands to talk long distance. The military also communicate in the HF and VHF bands.

3. Microwave Transmission

- Electromagnetic waves having frequencies between I and 300 GHz are called microwaves.
- Microwaves are unidirectional. When an antenna transmits microwave waves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned.
- The unidirectional property has an obvious advantage. A pair of antennas can be aligned without interfering with another pair of aligned antennas.

▪

The following describes some characteristics of microwave propagation:

- ➢ Microwave propagation is line-of-sight. Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall. The curvature of the earth as well as other blocking obstacles do not allow two short towers to communicate by using microwaves. Repeaters are often needed for long distance communication.
- ➢ Very high-frequency microwaves cannot penetrate walls. This characteristic can be a disadvantage if receivers are inside buildings.
- ➢ The microwave band is relatively wide, almost 299 GHz. Therefore wider sub-bands can be assigned, and a high data rate is possible
- ➢ Use of certain portions of the band requires permission from authorities.

- Microwave communication is so widely used for long-distance telephone communication, mobile phones, television distribution, and other purposes that a severe shortage of spectrum has developed.
- It has several key advantages over fiber. The main one is that no right of way is needed to lay down cables. By buying a small plot of ground every 50 km and putting a microwave tower on it, one can bypass the telephone system entirely.

4. Infrared Transmission

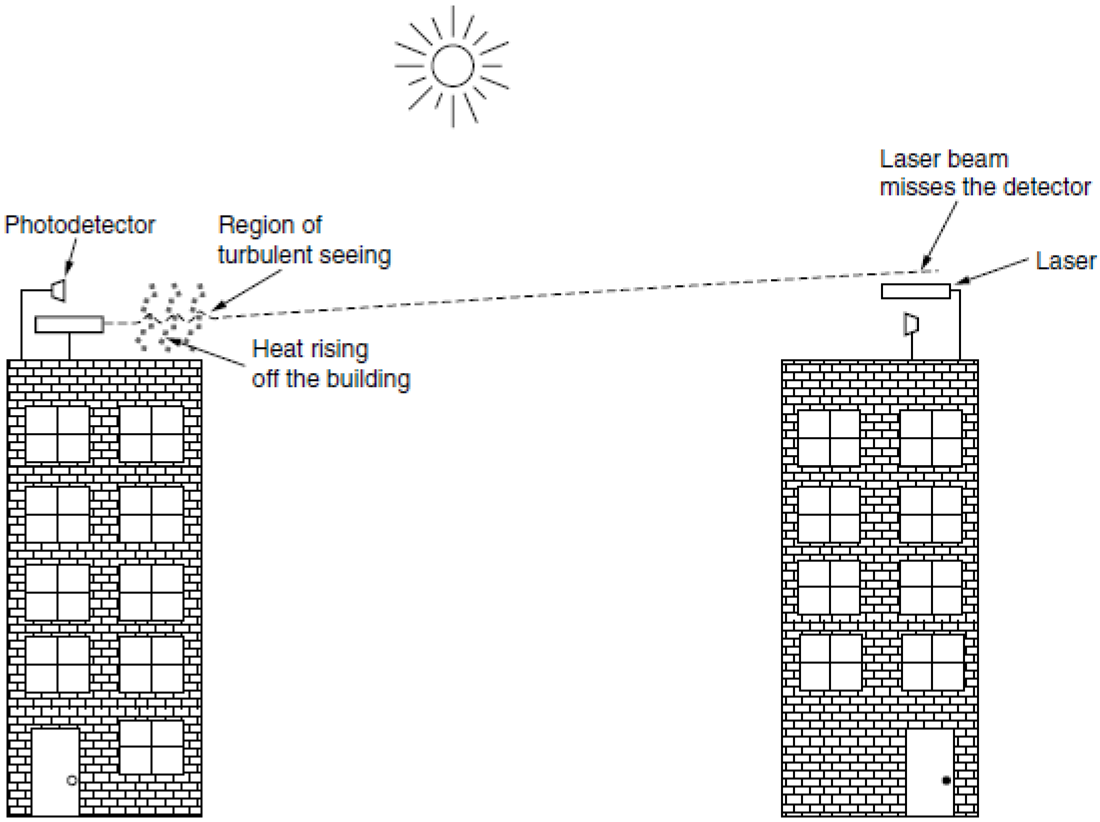
- Unguided infrared waves are widely used for short-range communication.
- Infrared waves, with frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm), can be used for short-range communication. Infrared waves, having high frequencies, cannot penetrate walls.
- The remote controls used for televisions, VCRs, and stereos all use infrared communication.
- They are relatively directional, cheap, and easy to build but have a major drawback: they do not pass through solid objects.
- In general, as we go from long-wave radio toward visible light, the waves behave more and more like light and less and less like radio.

- ➢ On the other hand, the fact that infrared waves do not pass through solid walls well is also a plus. It means that an infrared system in one room of a building will not interfere with a similar system in adjacent rooms or buildings: you cannot control your neighbor's television with your remote control.
- ➢ Security of infrared systems against eavesdropping is better than that of radio systems precisely for this reason. Therefore, no government license is needed to operate an infrared system, in contrast to radio systems, which must be licensed outside the ISM bands.
- ➢ Infrared communication has a limited use on the desktop, for example, to connect notebook computers and printers with the **IrDA (Infrared Data Association)** standard, but it is not a major player in the communication game.

5. Light Transmission

Unguided optical signaling or **free-space optics** has been in use for centuries. A more modern application is to connect the LANs in two buildings via lasers mounted on their roof-tops. Optical signaling using lasers is inherently unidirectional, so each end needs its own laser and its own photo-detector. This scheme offers very high bandwidth at very low cost and is relatively secure because it is difficult to tap a narrow laser beam. It is also relatively easy to install and, unlike microwave transmission, does not require an FCC license.

The organizers discovered the problem: heat from the sun during the daytime caused convection currents to rise up from the roof of the building, as shown in Figure.



Convection currents can interfere with laser communication systems. A bidirectional system with two lasers is pictured here.

Data communication can be layered on top of these displays by encoding information in the pattern at which LEDs turn on and off that is below the threshold of human perception.

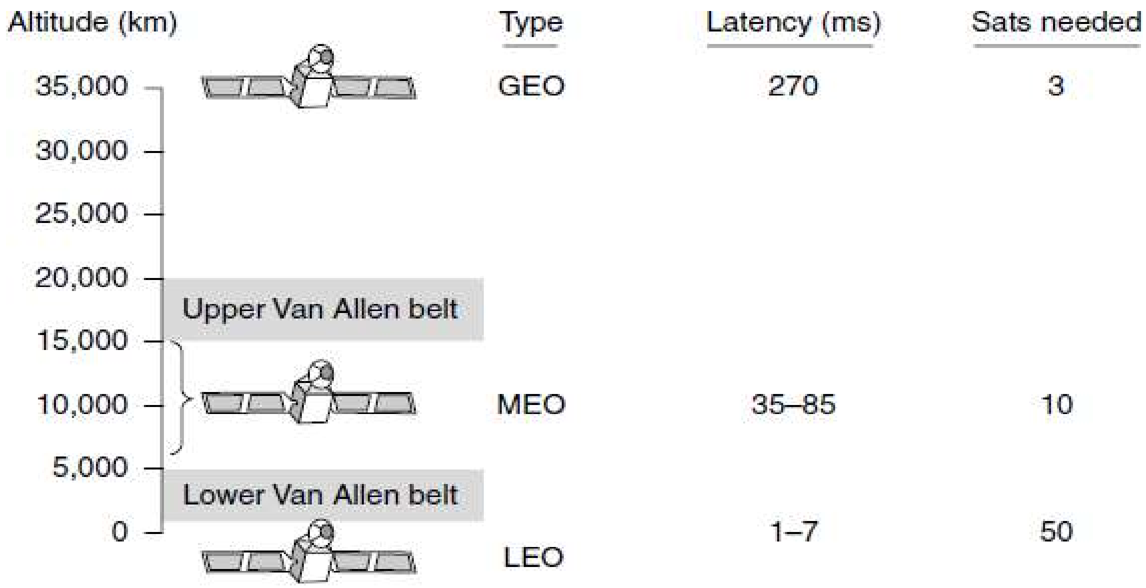
14. Explain the various communication satellites in detail? (11 MARKS) (NOV 2015)

- Communication satellites have some interesting properties that make them attractive for many applications. In its simplest form, a communication satellite can be thought of as a big microwave repeater in the sky.

- It contains several **transponders,** each of which listens to some portion of the spectrum, amplifies the incoming signal, and then rebroadcasts it at another frequency to avoid interference with the incoming signal. This mode of operation is known as a **bent pipe.**
- Digital processing can be added to separately manipulate or redirect data streams in the overall band, or digital information can even be received by the satellite and rebroadcast.
- Regenerating signals in this way improves performance compared to a bent pipe because the satellite does not amplify noise in the upward signal. The downward beams can be broad, covering a substantial fraction of the earth's surface, or narrow, covering an area only hundreds of kilometers in diameter.

- According to **Kepler's law**, the orbital period of a satellite varies as the radius of the orbit to the 3/2 power. The higher the satellite, the longer the period. Near the surface of the earth, the period is about 90 minutes.
- Consequently, low-orbit satellites pass out of view fairly quickly, so many of them are needed to provide continuous coverage and ground antennas must track them.
- At an altitude of about 35,800 km, the period is 24 hours. At an altitude of 384,000 km, the period is about one month, as anyone who has observed the moon regularly can testify.

Another issue is the presence of the Van Allen belts, layers of highly charged particles trapped by the earth's magnetic field. Any satellite flying within them would be destroyed fairly quickly by the particles. These factors lead to three regions in which satellites can be placed safely.



Communication satellites and some of their properties, including altitude above the earth, round-trip delay time, and number of satellites needed for global coverage.

There is only one orbit, at an altitude of 35,786 km for the GEO satellite. MEO satellites are located at altitudes between 5000 and 15,000 km. LEO satellites are normally below an altitude of 2000 km.

The various communication satellites are

4. Geostationary Satellites (GEO)
5. Medium-Earth Orbit Satellites (MEO)
6. Low-Earth Orbit Satellites (LEO)

1. Geostationary Satellites

- In 1945, the science fiction writer Arthur C. Clarke calculated that a satellite at an altitude of 35,800 km in a circular equatorial orbit would appear to remain motionless in the sky, so it would not need to be tracked.
- Geostationary satellites, including the orbits, solar panels, radio frequencies, and launch procedures.
- The invention of the transistor changed all that, and the first artificial communication satellite, Telstar, was launched in July 1962. Since then, communication satellites have become a multibillion dollar business and the only aspect of outer space that has become highly profitable. These high-flying satellites are often called **GEO (Geostationary Earth Orbit)** satellites.
- Modern satellites can be quite large, weighing over 5000 kg and consuming several kilowatts of electric power produced by the solar panels. The effects of solar, lunar, and planetary gravity tend to move them away from their assigned orbit slots and orientations, an effect countered by on-board rocket motors. This fine-tuning activity is called **station keeping.**

The principal satellite bands

Band	Downlink	Uplink	Bandwidth	Problems
L	1.5 GHz	1.6 GHz	15 MHz	Low bandwidth; crowded
S	1.9 GHz	2.2 GHz	70 MHz	Low bandwidth; crowded
C	4.0 GHz	6.0 GHz	500 MHz	Terrestrial interference
Ku	11 GHz	14 GHz	500 MHz	Rain
Ka	20 GHz	30 GHz	3500 MHz	Rain, equipment cost

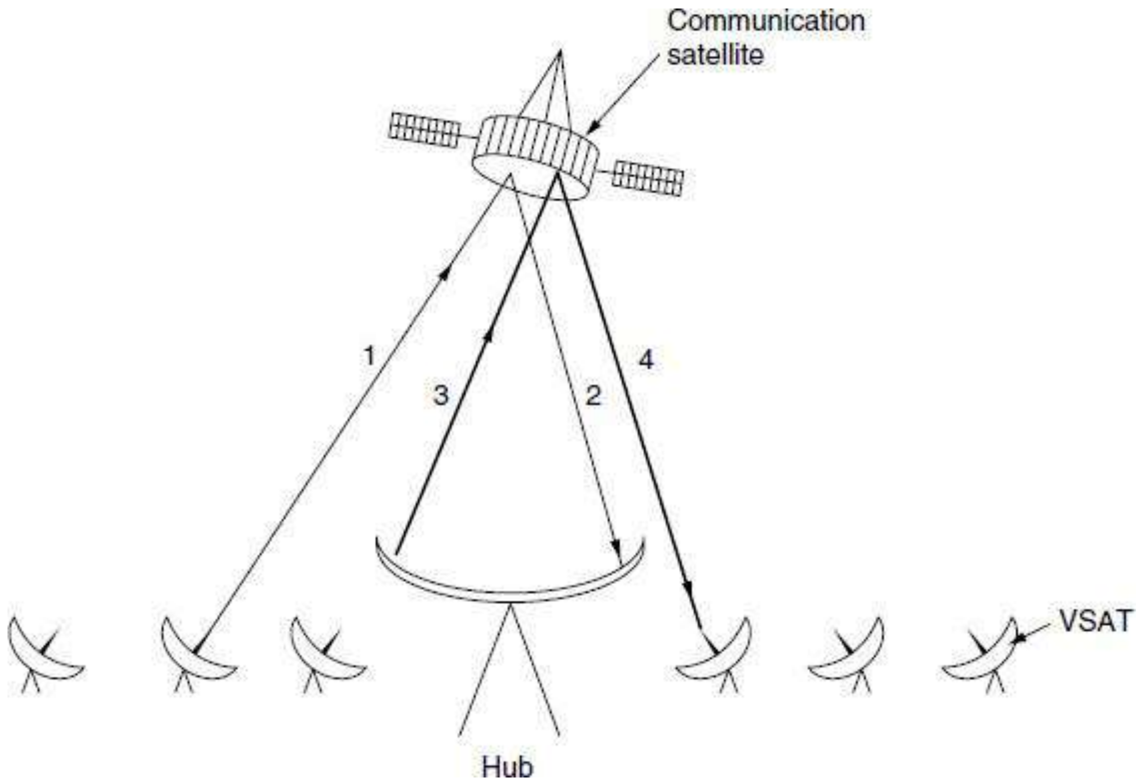
A modern satellite has around 40 transponders, most often with a 36-MHz bandwidth. Usually, each transponder operates as a bent pipe, but recent satellites have some on-board processing capacity, allowing more sophisticated operation.

The first geostationary satellites had a single spatial beam that illuminated about 1/3 of the earth's surface, called its **footprint.** With the enormous decline in the price, size, and power requirements of microelectronics, a much more sophisticated broadcasting strategy has become possible. Each satellite is equipped with multiple antennas and multiple transponders.

Each downward beam can be focused on a small geographical area, so multiple upward and downward transmissions can take place simultaneously. Typically, these so-called **spot beams** are elliptically shaped, and can be as small as a few hundred km in diameter.

A recent development in the communication satellite world is the development of low-cost micro-stations, sometimes called **VSATs (Very Small Aperture Terminals).** The terminals have 1-meter or smaller antennas (versus 10 m for a standard GEO antenna) and can put out about 1 watt of power. The uplink is generally good for up to 1 Mbps, but the downlink is often up to several megabits/sec. Direct broadcast satellite television uses this technology for one-way transmission.

In many VSAT systems, the micro-stations do not have enough power to communicate directly with one another (via the satellite, of course). Instead, a special ground station, the **hub,** with a large, high-gain antenna is needed to relay traffic between VSATs, as shown in Figure. In this mode of operation, either the sender or the receiver has a large antenna and a powerful amplifier. The trade-off is a longer delay in return for having cheaper end-user stations.



VSATs using a hub

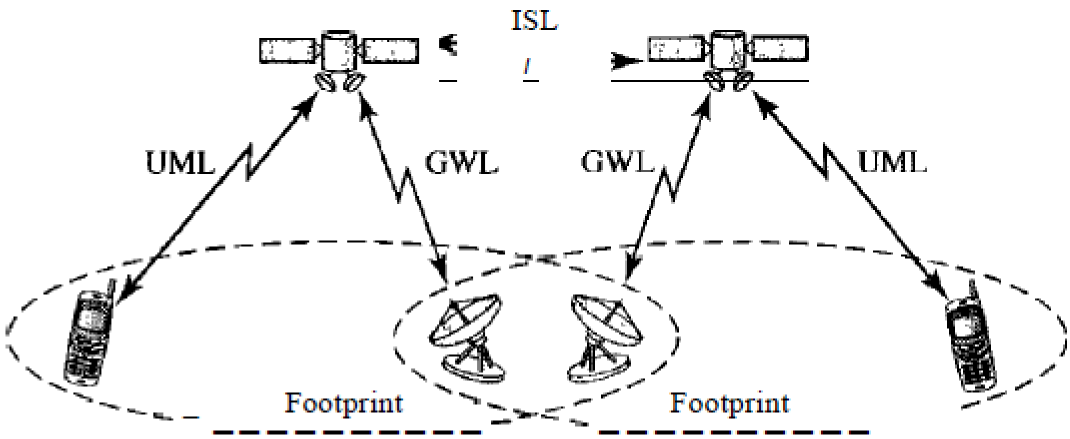
2. Medium-Earth Orbit Satellites

- At much lower altitudes, between the two Van Allen belts, we find the MEO (Medium-Earth Orbit) satellites.
- As viewed from the earth, these drift slowly in longitude, taking something like 6 hours to circle the earth.
- Accordingly, they must be tracked as they move through the sky. Because they are lower than the GEOs, they have a smaller footprint on the ground and require less powerful transmitters to reach them.
- Currently they are used for navigation systems rather than telecommunications.
- The constellation of roughly 30 GPS (Global Positioning System) satellites orbiting at about 20,200 km are examples of MEO satellites.

- GPS is used by military forces.

3. Low-Earth Orbit Satellites

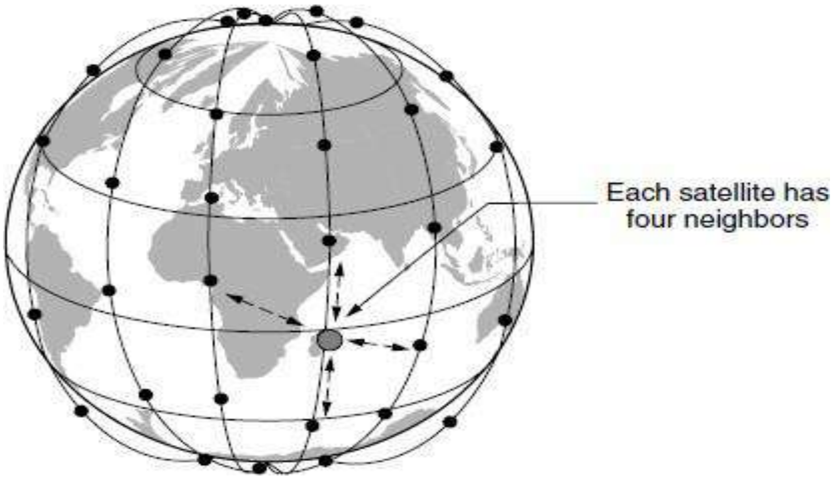
- Low-Earth-orbit (LEO) satellites have polar orbits. The altitude is between 500 and 2000 km, with a rotation period of 90 to 120 min.
- The satellite has a speed of 20,000 to 25,000 km/h.
- An LEO system usually has a cellular type of access, similar to the cellular telephone system.
- The footprint normally has a diameter of 8000 km. Because LEO satellites are close to Earth, the round-trip time propagation delay is normally less than 20 ms, which is acceptable for audio communication.
- An LEO system is made of a constellation of satellites that work together as a network; each satellite acts as a switch.
- Satellites that are close to each other are connected through inter satellite links (ISLs).
- A mobile system communicates with the satellite through a user mobile link (UML).
- A satellite can also communicate with an Earth station (gateway) through a gateway link (GWL).



LEO satellite network

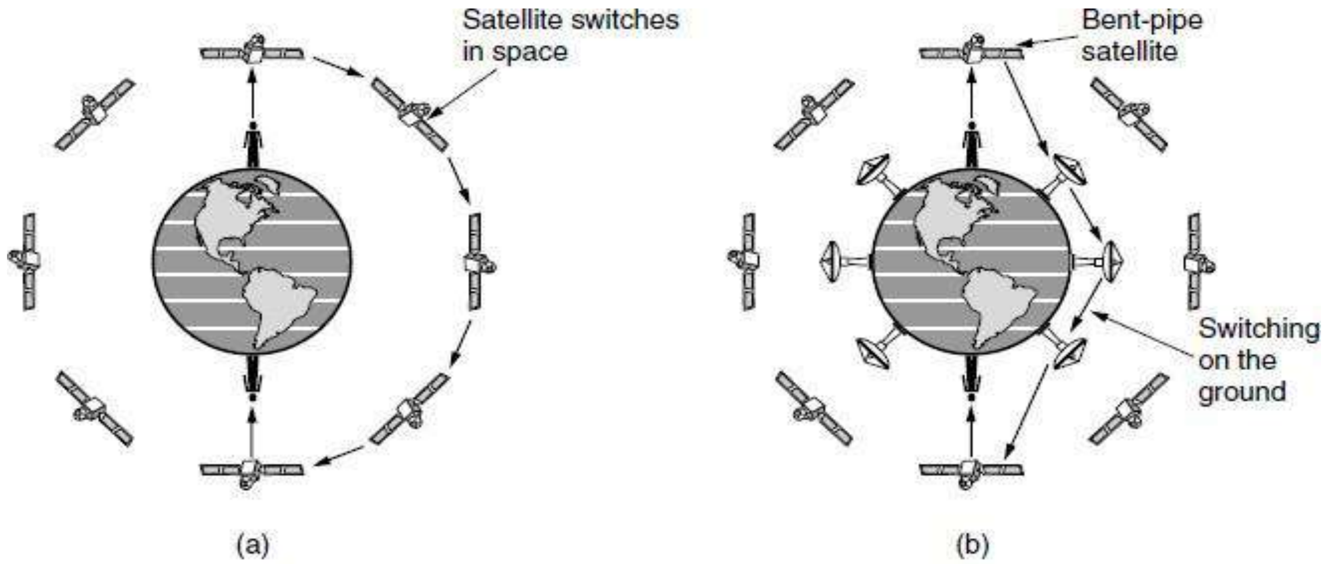
The **Iridium** service provides voice, data, paging, fax, and navigation service everywhere on land, air, and sea, via hand-held devices that communicate directly with the Iridium satellites. Customers include the maritime, aviation, and oil exploration industries, as well as people traveling in parts of the world lacking a telecom infrastructure.

The Iridium satellites are positioned at an altitude of 750 km, in circular polar orbits. They are arranged in north-south necklaces, with one satellite every 32 degrees of latitude, as shown in Figure. Each satellite has a maximum of 48 cells (spot beams) and a capacity of 3840 channels, some of which are used for paging and navigation, while others are used for data and voice.



The Iridium satellites form six necklaces around the earth

An interesting property of Iridium is that communication between distant customers takes place in space, as shown in Figure.



(a) Relaying in space. (b) Relaying on the ground.

An alternative design to Iridium is **Globalstar**. It is based on 48 LEO satellites but uses a different switching scheme than that of Iridium. Whereas Iridium relays calls from satellite to satellite, which requires sophisticated switching equipment in the satellites, Globalstar uses a traditional bent-pipe design.

15. Explain in details the different topologies of networks with examples? (11 MARKS) (NOV 2016)

Physical Topology

- The term physical topology refers to the way in which a network is laid out physically: Two or more devices connect to a link; two or more links form a topology.
- The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.

Categories of topology

There are four basic topologies possible:

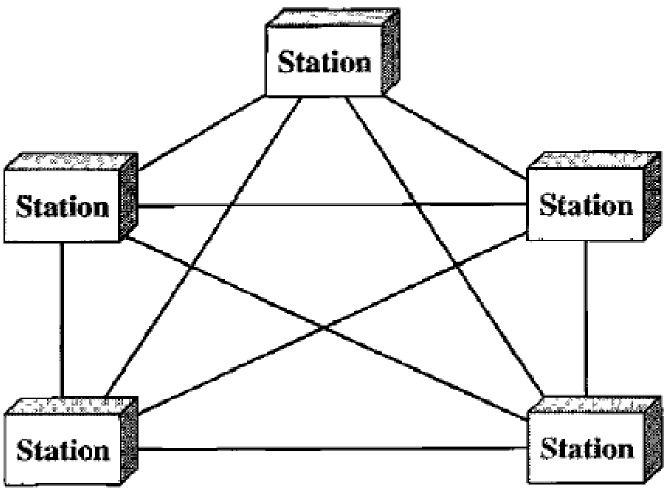
1. Mesh
2. Star
3. Bus and
4. Ring

1. Mesh topology

- In a mesh topology, every device has a dedicated point-to-point link to every other device.
- The term dedicated means that the link carries traffic only between the two devices it connects.
- To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node.
- Node 1 must be connected to n - I nodes, node 2 must be connected to n – 1 nodes, and finally node n must be connected to n - 1 nodes.
- We need n(n - 1) physical links. However, if each physical link allows communication in both directions (duplex mode), we can divide the number of links by 2.
- In other words, we can say that in a mesh topology, we need

$$n(n-1)/2$$

To accommodate that many links, every device on the network must have n – 1 input/output (I/O) ports (Figure) to be connected to the other n - 1 stations.



A fully connected mesh topology (five devices)

Advantages

A mesh offers several advantages over other network topologies:

1. First, the use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
2. Second, a mesh topology is robust. If one link becomes unusable, it does not incapacitate the entire system.

3. Third, there is the advantage of privacy or security. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages. Finally, point-to-point links make fault identification and fault isolation easy.
4. Traffic can be routed to avoid links with suspected problems. This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.

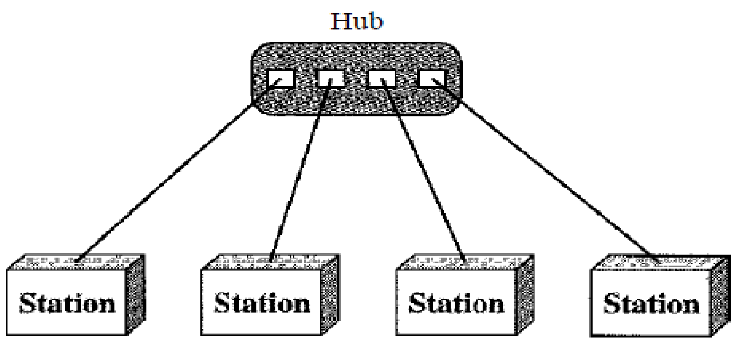
Disadvantages

1. The main disadvantages of a mesh are related to the amount of cabling and the number of I/O ports required.
2. First, because every device must be connected to every other device, installation and reconnection are difficult.
3. Second, the sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate.
4. Finally, the hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

One practical example of a mesh topology is the connection of telephone regional offices in which each regional office needs to be connected to every other regional office.

2. Star topology

- In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub.
- The devices are not directly linked to one another.
- Unlike a mesh topology, a star topology does not allow direct traffic between devices.
- The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.



A star topology connecting four stations

- A star topology is less expensive than a mesh topology.
- In a star, each device needs only one link and one I/O port to connect it to any number of others. This factor also makes it easy to install and reconfigure.
- Far less cabling needs to be housed, and additions, moves, and deletions involve only one connection: between that device and the hub.

Advantages

Other advantages include robustness.

1. If one link fails, only that link is affected. All other links remain active.
2. This factor also lends itself to easy fault identification and fault isolation.
3. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

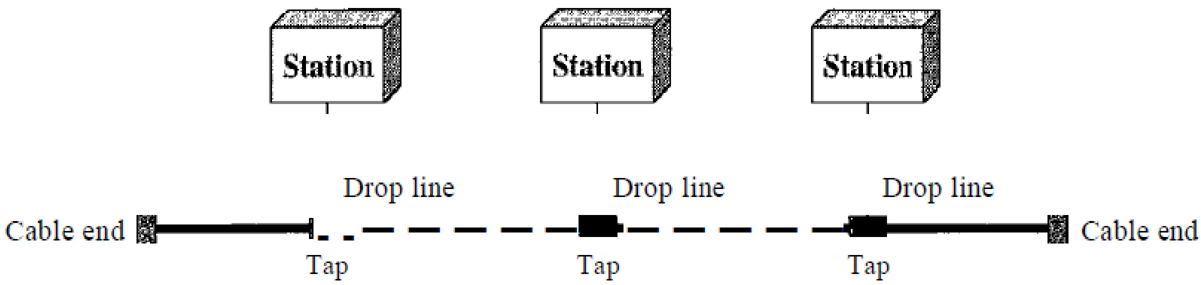
Disadvantages

1. One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.
2. Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies (such as ring or bus).

The star topology is used in local-area networks (LANs). High-speed LANs often use a star topology with a central hub.

3. Bus Topology

Bus topology, on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in a network.



A bus topology connecting three stations

Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable. A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core. As a signal travels along the backbone, some of its energy is transformed into heat. Therefore, it becomes weaker and weaker as it travels farther and farther. For this reason there is a limit on the number of taps a bus can support and on the distance between those taps.

Advantages

1. Advantages of a bus topology include ease of installation.
2. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies.
3. In a star, for example, four network devices in the same room require four lengths of cable reaching all the way to the hub. In a bus, this redundancy is eliminated.

4. Only the backbone cable stretches through the entire facility. Each drop line has to reach only as far as the nearest point on the backbone.

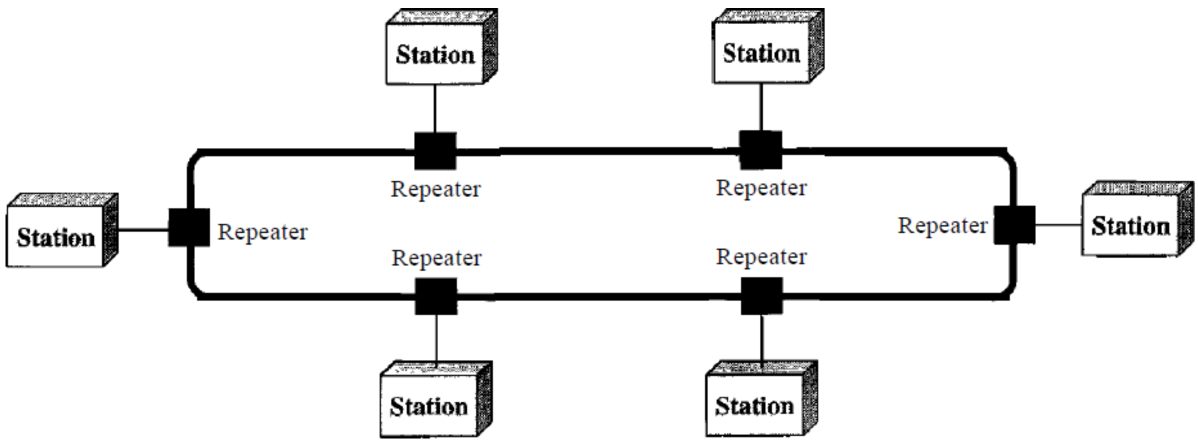
Disadvantages

1. Disadvantages include difficult reconnection and fault isolation.
2. A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices.
3. Signal reflection at the taps can cause degradation in quality. This degradation can be controlled by limiting the number and spacing of devices connected to a given length of cable.
4. Adding new devices may therefore require modification or replacement of the backbone.
5. In addition, a fault or break in the bus cable stops all transmission, even between devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

Bus topology was the one of the first topologies used in the design of early local area networks. Ethernet LANs can use a bus topology.

4. Ring Topology

- In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- Each device in the ring incorporates a repeater.
- When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.



A ring topology connecting six stations

Advantages

1. A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbors (either physically or logically).
2. To add or delete a device requires changing only two connections. The only constraints are media and traffic considerations.
3. In addition, fault isolation is simplified.

4. Generally in a ring, a signal is circulating at all times. If one device does not receive a signal within a specified period, it can issue an alarm. The alarm alerts the network operator to the problem and its location.

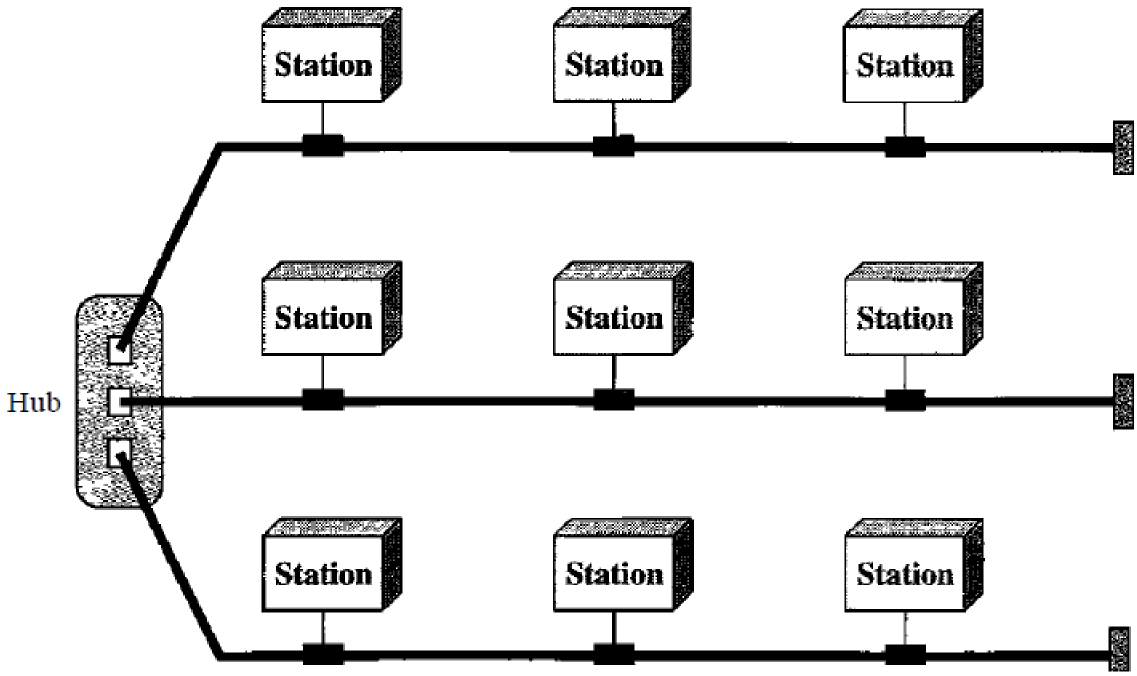
Disadvantages

However, unidirectional traffic can be a disadvantage. In a simple ring, a break in the ring (such as a disabled station) can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break.

Ring topology was prevalent when IBM introduced its local-area network Token Ring. Today, the need for higher-speed LANs has made this topology less popular.

Hybrid Topology

A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology**.**



A hybrid topology: a star backbone with three bus networks

PONDICHERRY UNIVERSITY QUESTIONS

2 MARKS

1. What are the functions of the transport layer in the OSI reference model? **(NOV 2015) (Ref.Qn.No.35, Pg.no.11)**
2. What are the standards used in wireless communication? **(NOV 2015) (Ref.Qn.No.59, Pg.no.16)**
3. Name the two types of transmission technology. **(MAY 2016, 2017) (Ref.Qn.No.15, Pg.no.7)**
4. What is the difference between a passive star and an active repeater in a fiber optic network? **(MAY 2016) (Ref.Qn.No.57, Pg.no.15)**
5. Differentiate guided and unguided transmission medium. **(NOV 2016) (Ref.Qn.No.50, Pg.no.14)**
6. What is the purpose of network interface card? **(NOV 2016) (Ref.Qn.No.21, Pg.no.9)**
7. Compare the Connection-oriented and Connectionless service. **(MAY 2017) (Ref.Qn.No.30, Pg.no.10)**

11 MARKS

NOV 2015 (REGULAR)

1. (a) Discuss a different types of transmission media used in network. (6) **(Ref.Qn.No.12, Pg.no.60)**
   (b) Write a brief note on communication satellite. (5) **(Ref.Qn.No.14, Pg.no.72)**

(OR)

2. (a) Compare the internet and OSI reference models and list the advantages of each. (6) **(Ref.Qn.No.7, Pg.no.38)**
   (b) Briefly discuss the theoretical basis behind communication system. (5) **(Ref.Qn.No.11, Pg.no.57)**

MAY 2016 (ARREAR)

1. Explain detail about the OSI reference model of computer network architecture. **(Ref.Qn.No.7, Pg.no.38)**

(OR)

2. Explain briefly about the fiber optics. **(Ref.Qn.No.12, Pg.no.63)**

NOV 2016 (REGULAR)

1. Describe briefly the various layers and functions of OSI model. **(Ref.Qn.No.7, Pg.no.38)**

(OR)

2. Explain in details the different topologies of networks with examples. **(Ref.Qn.No.15, Pg.no.77)**

MAY 2017 (ARREAR)

1. Explain briefly the different types of guided transmission media and unguided transmission media used for data transmission. **(Ref.Qn.No.12,13 Pg.no.60,66)**

(OR)

2. Compare and contrast the ISO/OSI reference model with the TCP/IP reference model. **(Ref.Qn.No.9, Pg.no.4)**